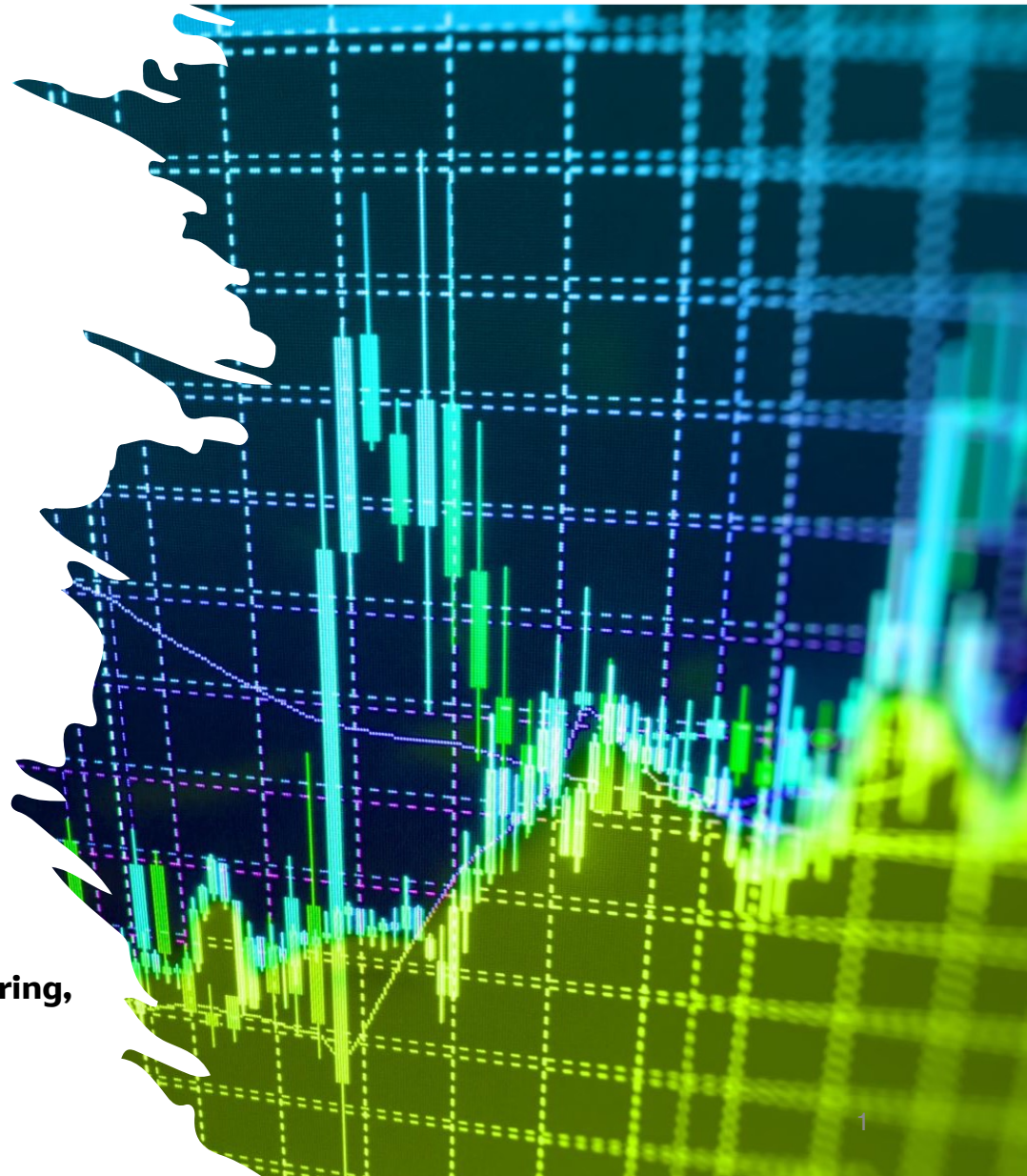


Introduction to AI

Prof. Peerapon Vateekul, Ph.D.

**Department of Computer Engineering, Faculty of Engineering,
Chulalongkorn University**

Peerapon.v@chula.ac.th





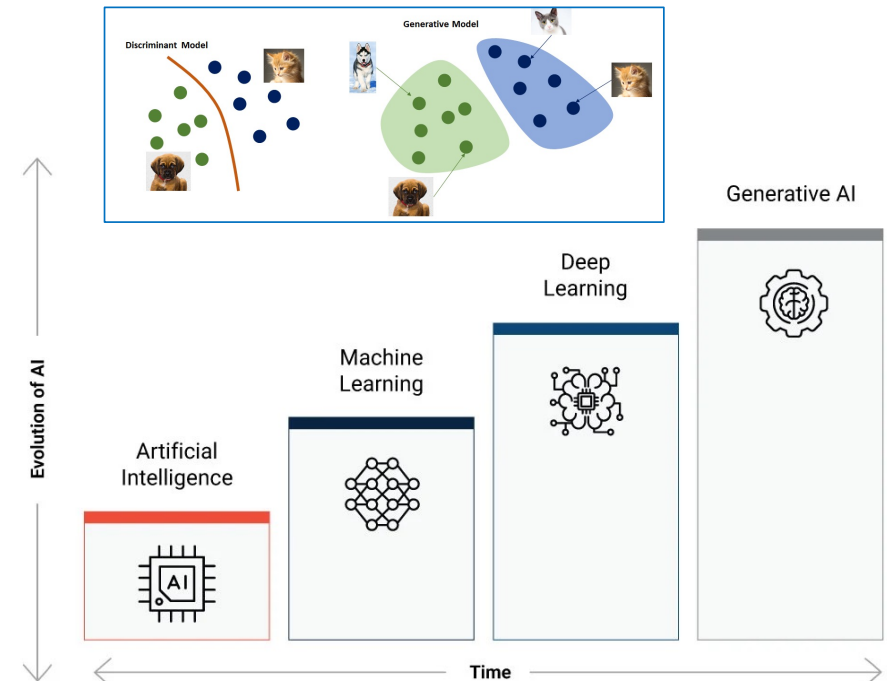
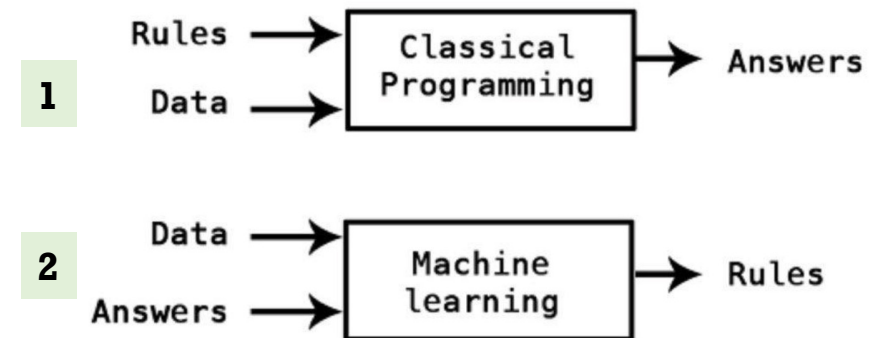
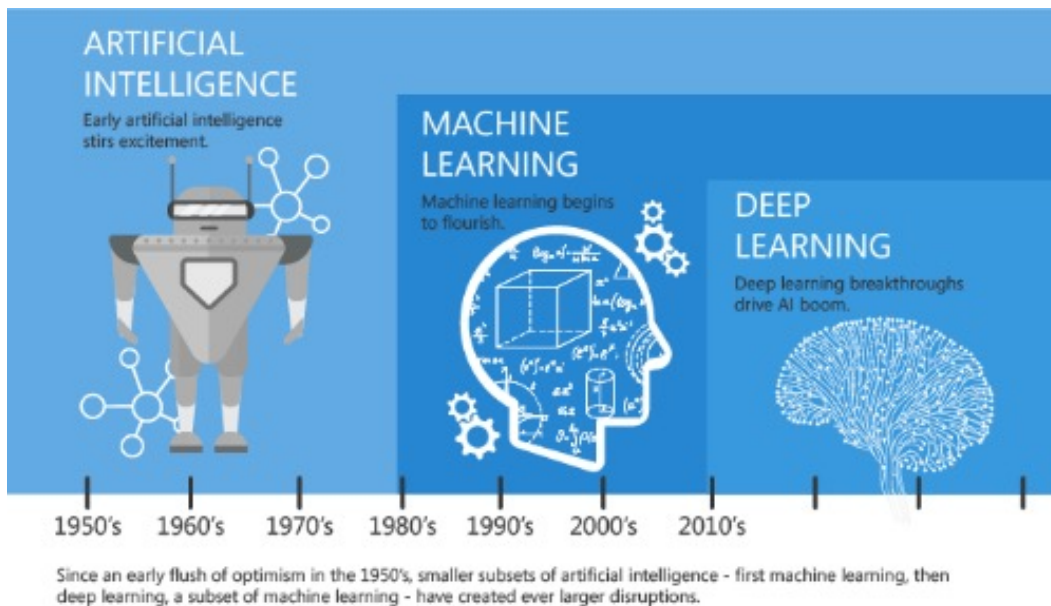
AI/ML



IRIS

AI = Automation

- 1) Rule-based AI
- 2) Machine Learning (ML)



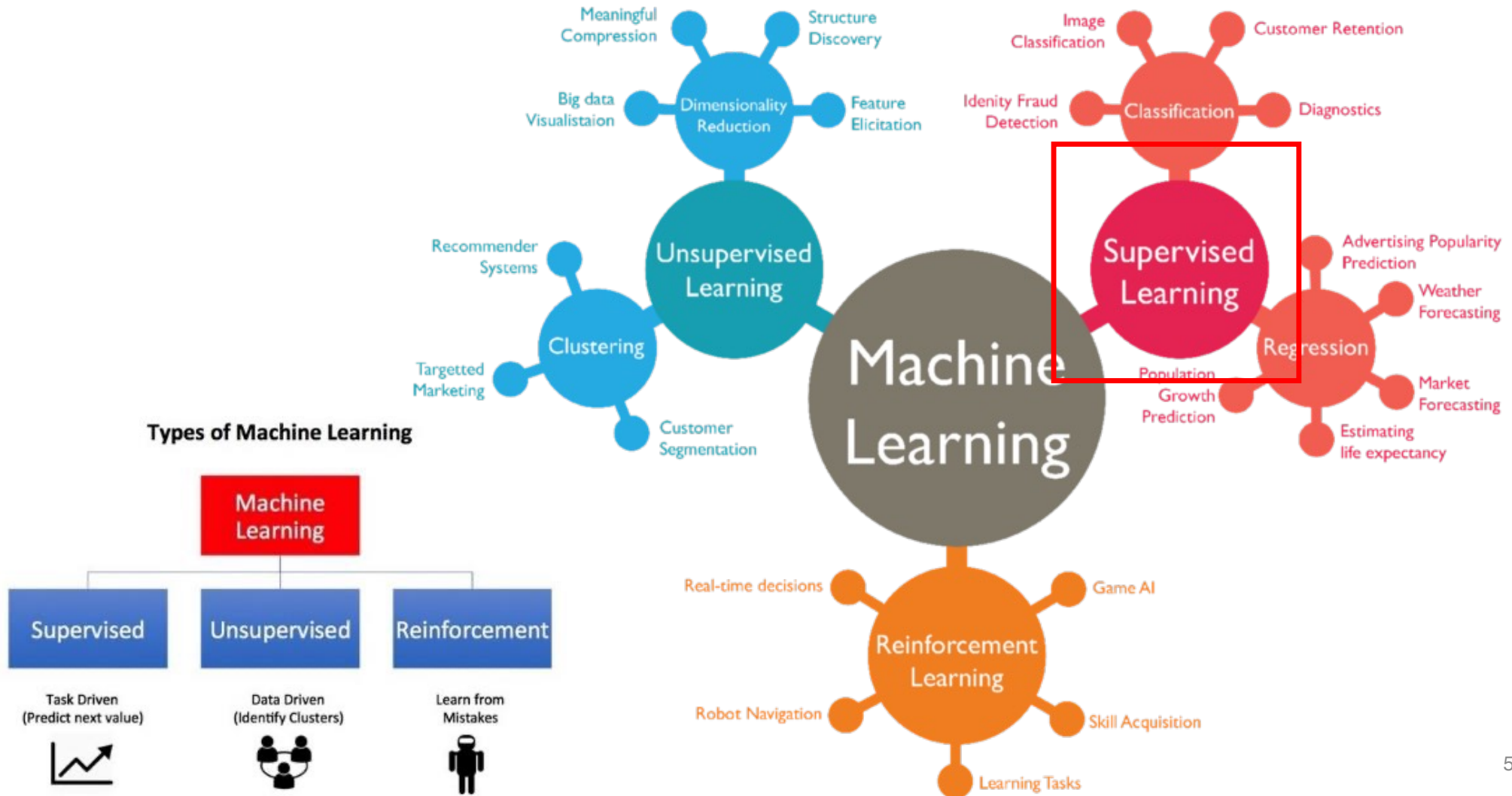
<https://mc.ai/machine-learning-basics-artificial-intelligence-machine-learning-and-deep-learning/>³

Terminology: Data table

inputs				target
Age	Income	Gender	Province	Purchase
25	25,000	Female	Bangkok	Yes
35	50,000	Female	Nontaburi	Yes
32	35,000	Male	Bangkok	No

- Row
 - Example, instance, case, observation, subject
- Column
 - Feature, variable, attribute
- Input
 - Predictor, independent, explanatory variable
- Target
 - Output, outcome, response, dependent variable

Task1: Supervised learning (predictive task)



Task1: Supervised learning (predictive task)

Handcrafted features



Training Data



inputs				target
Age	Income	Gender	Province	Purchase
25	25,000	Female	Bangkok	Yes
35	50,000	Female	Nontaburi	Yes
32	35,000	Male	Bangkok	No

Testing Data

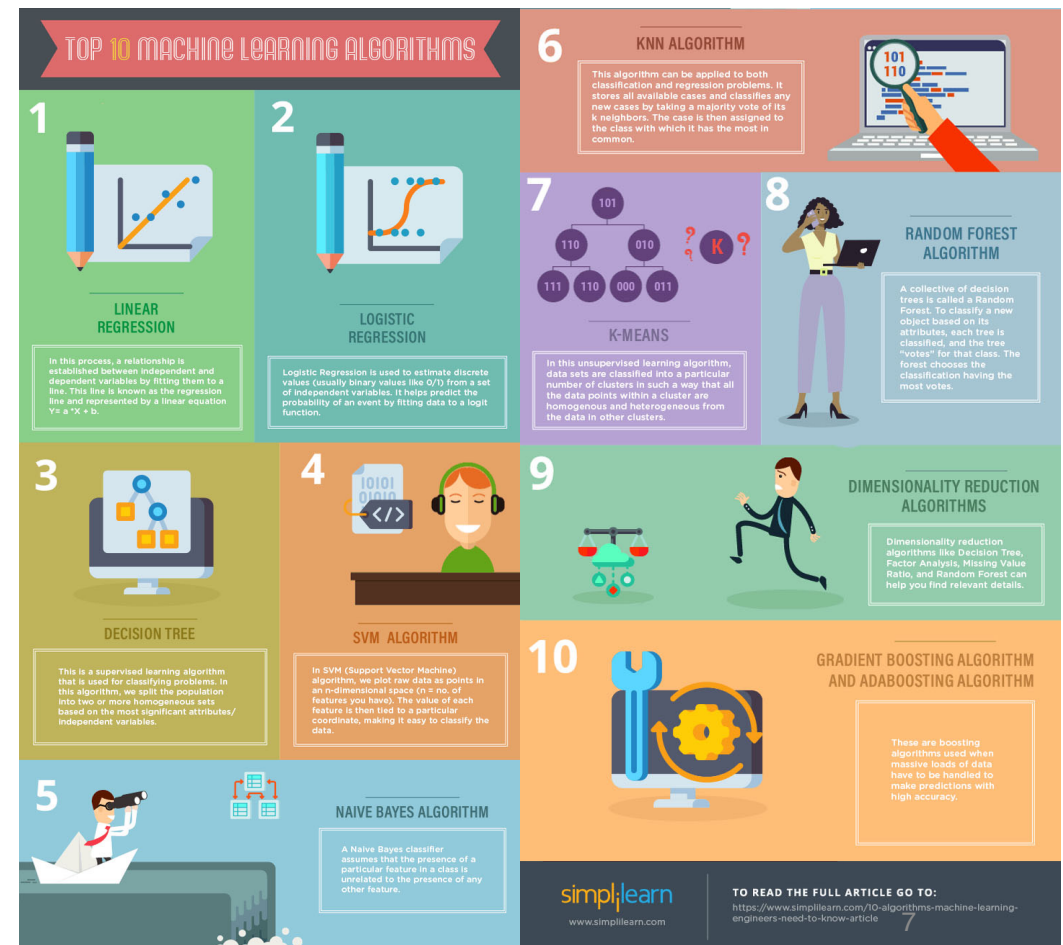


Age	Income	Gender	Province	Purchase
25	25,000	Female	Bangkok	?

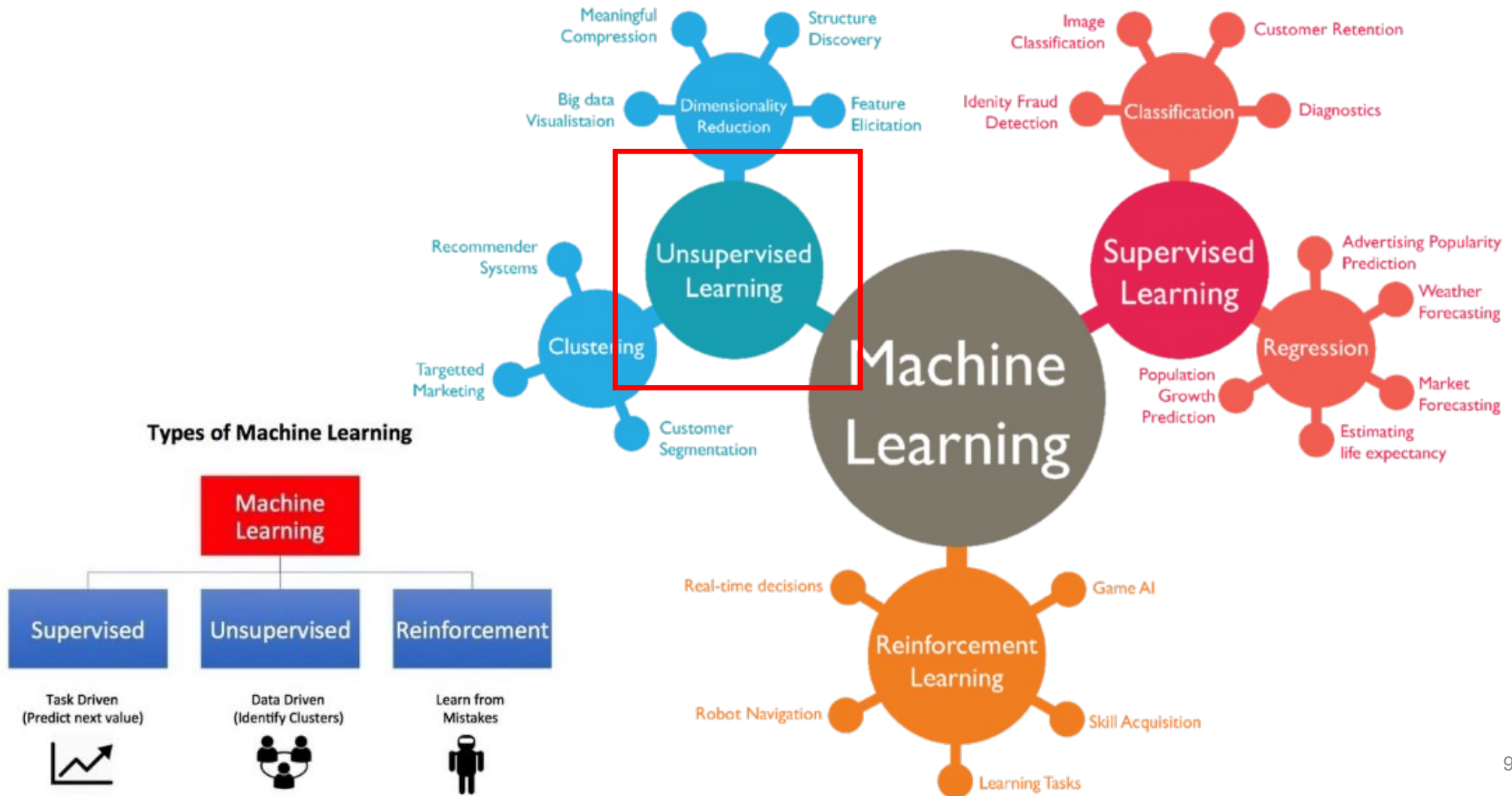
Application: Direct Target Marketing

Prediction Algorithms

- Decision Tree
- (Logistic) Regression
- Neural Networks (NN)
- k-Nearest Neighbors (kNN)
- Support Vector Machine
- Deep Learning



Task2: Unsupervised learning (descriptive task)



Task2: Unsupervised learning (descriptive task)

Training Data



inputs				target
Age	Income	Gender	Province	Purchase
25	25,000	Female	Bangkok	Yes
35	50,000	Female	Nontaburi	Yes
32	35,000	Male	Bangkok	No



Testing Data



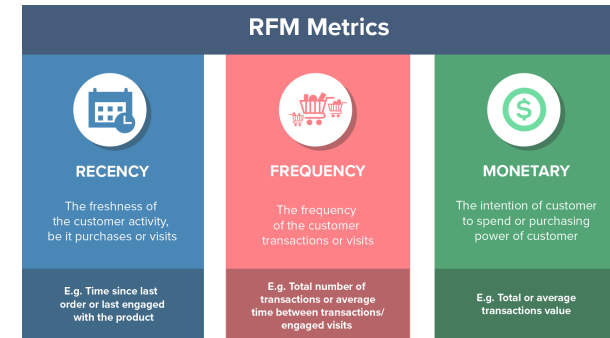
Age	Income	Gender	Province	Purchase
25	25,000	Female	Bangkok	?



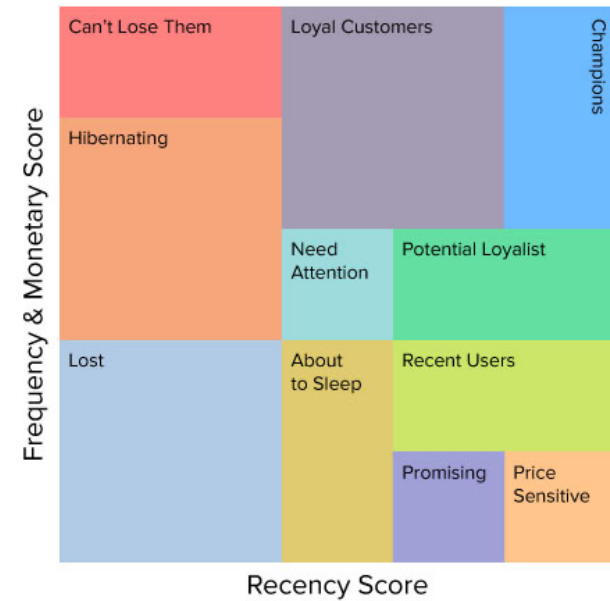
Clustering



IRIS



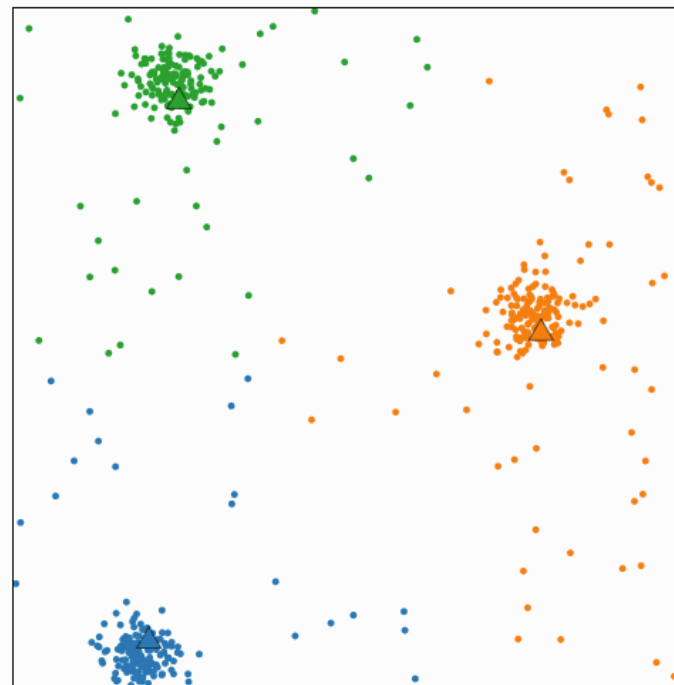
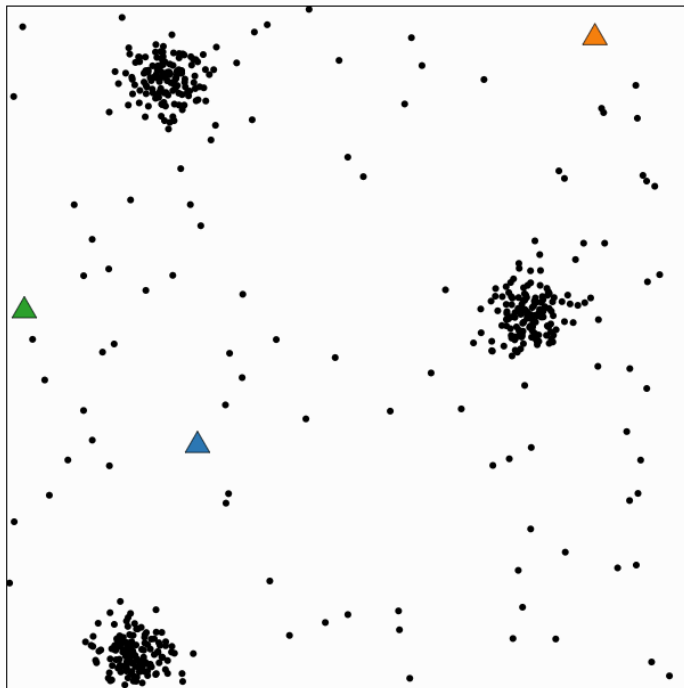
RFM Analysis



K-means Clustering

<http://web.stanford.edu/class/ee103/visualizations/kmeans/kmeans.html>

Visualizing K-Means Clustering



Mean square point-centroid distance: 6191.49

The k -means algorithm is an iterative method for clustering a set of N points (vectors) into k groups or clusters of points.

Algorithm

Repeat until convergence:

Find closest centroid

Find the closest centroid to each point, and group points that share the same closest centroid.

Update centroid

Update each centroid to be the mean of the points in its group.

Update centroid

Data

Clustered points ☒ Random

Number of clusters : 3

Number of centroids: 3

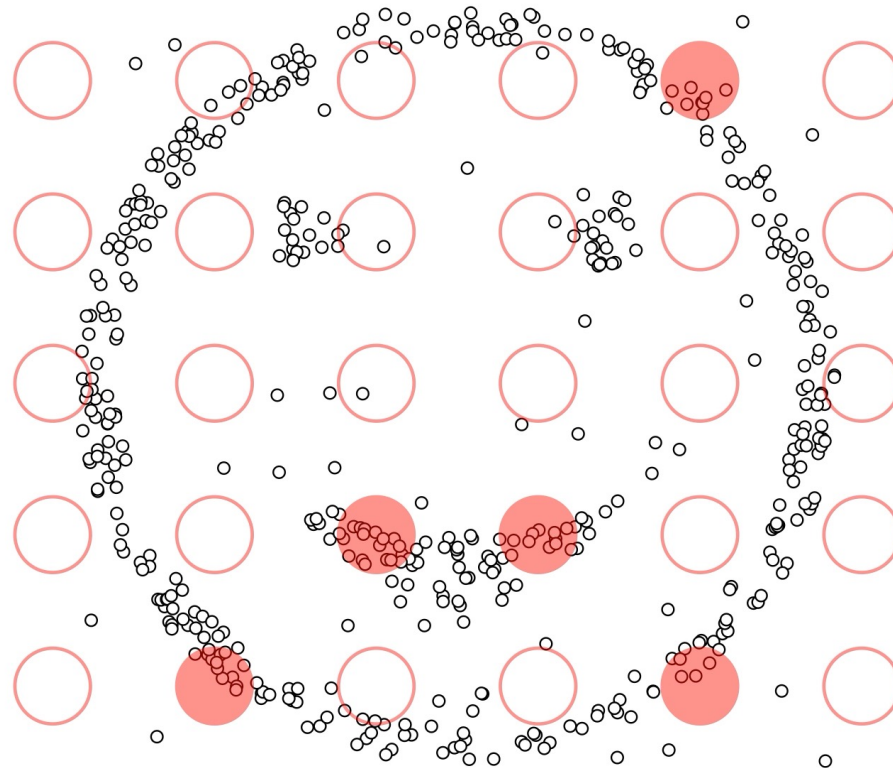
New points

New centroids

IRIS

DBSCAN

<https://www.naftaliharris.com/blog/visualizing-dbscan-clustering/>



epsilon = 1.00
minPoints = 4

IRIS

Restart



GO!



HomeInstallationDocumentationExamples

Previous

Next

Up

scikit-learn v0.21.3

Other versions

Please cite us if you use the software.

sklearn.cluster.DBSCAN

Examples using sklearn.cluster.DBSCAN

Google Custom Search

Q

sklearn.cluster.DBSCAN

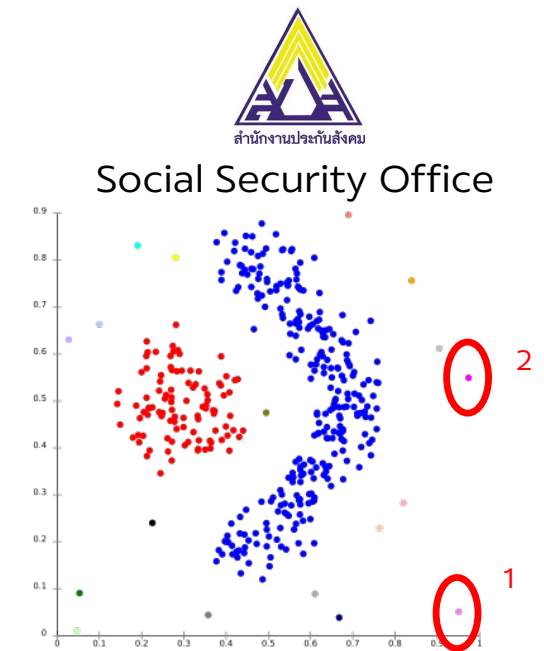
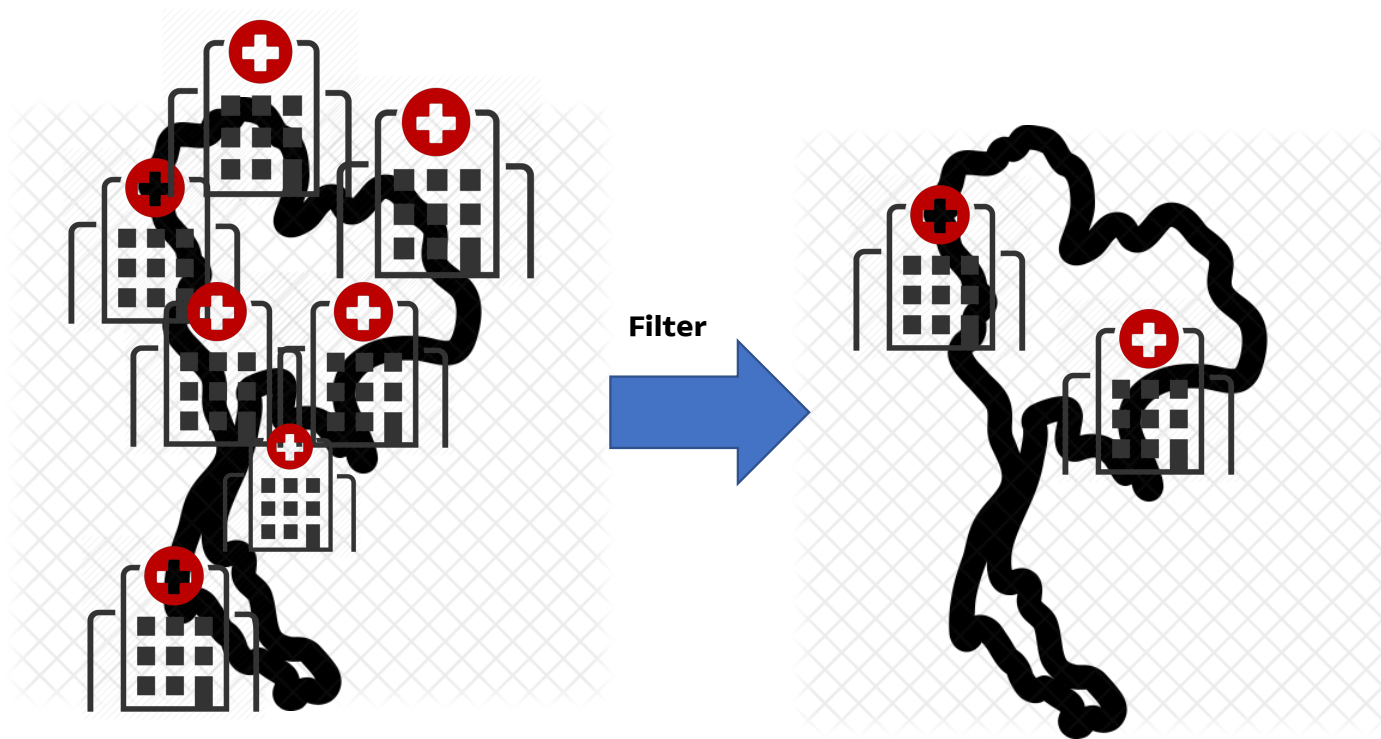
```
class sklearn.cluster.DBSCAN (eps=0.5, min_samples=5, metric='euclidean', metric_params=None, algorithm='auto', leaf_size=30, p=None, n_jobs=None)
```

[\[source\]](#)

Perform DBSCAN clustering from vector array or distance matrix.

DBSCAN - Density-Based Spatial Clustering of Applications with Noise. Finds core samples of high density and expands clusters from them. Good for data which contains clusters of similar density.

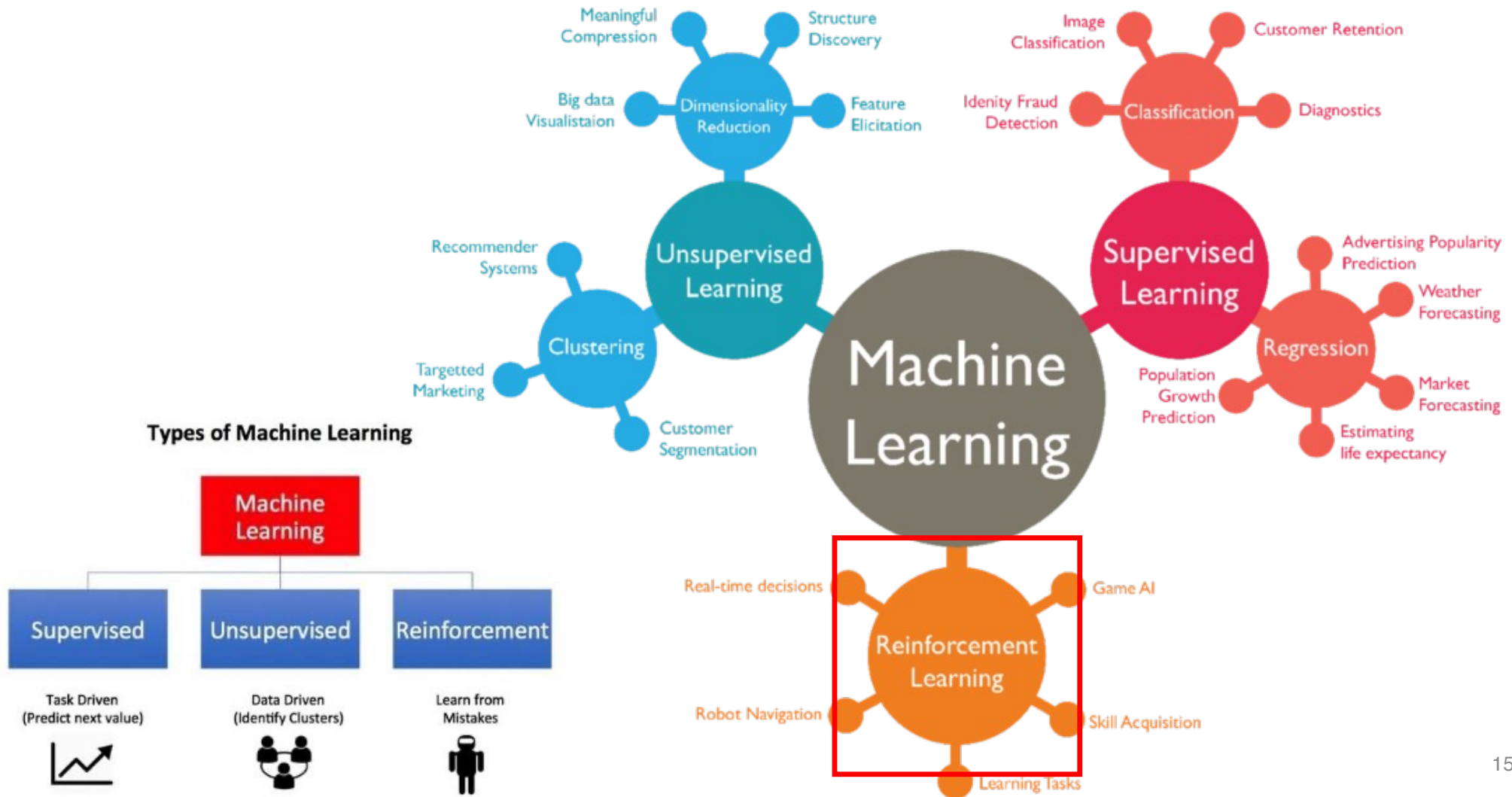
Fraud Detection (Unsupervised Learning)



IRIS

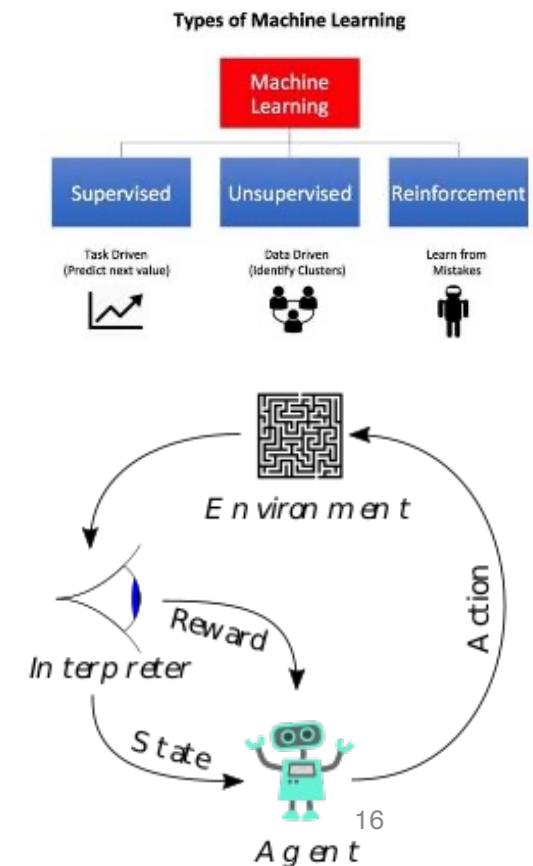
บอกได้ว่าควรตรวจสอบ รพ. ใด และควรตรวจสอบไล่จากสาเหตุใด

Task3: Reinforcement learning (optimization task)



Task3: Reinforcement Learning (cont.)

<https://www.youtube.com/watch?v=fiQsmdwEGT8>



Reinforcement Learning (cont.)

<https://www.zdnet.com/article/deepmind-alphago-zero-learns-on-its-own-without-meatbag-intervention/>

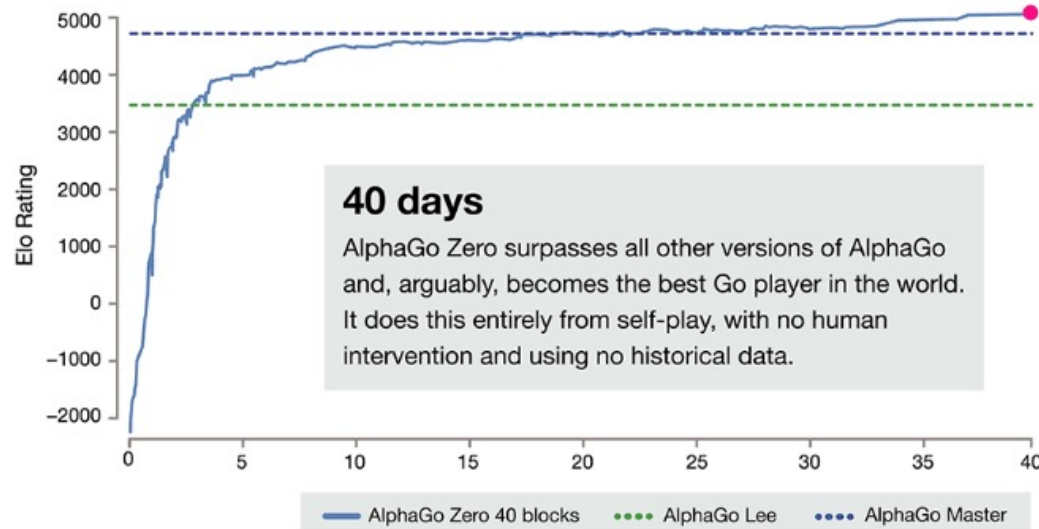
DeepMind AlphaGo Zero learns on its own without meatbag intervention

The latest iteration of DeepMind's Go-playing AI has taught itself and bested other versions of AlphaGo.

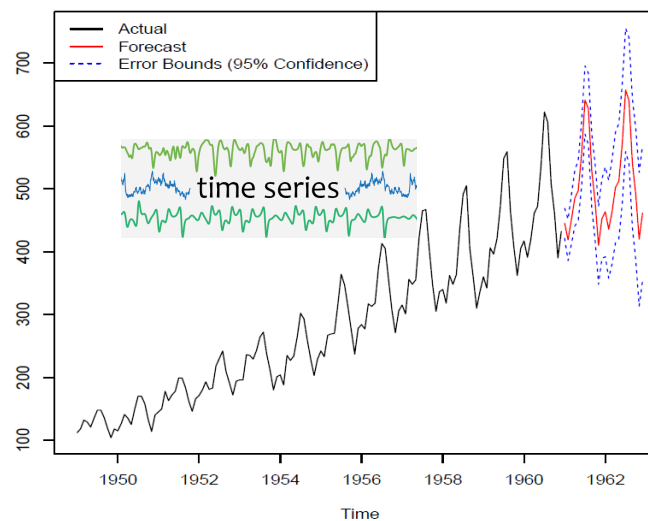
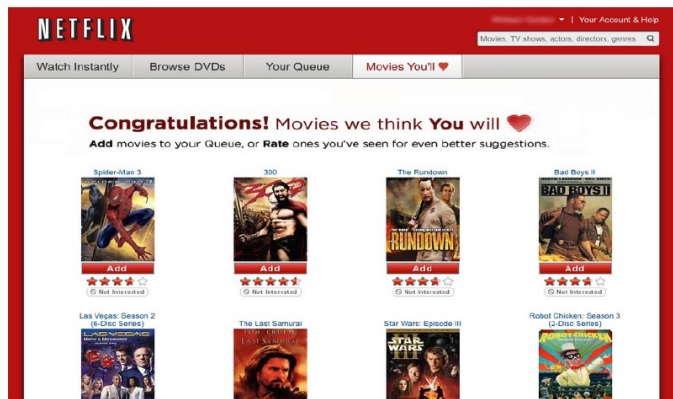
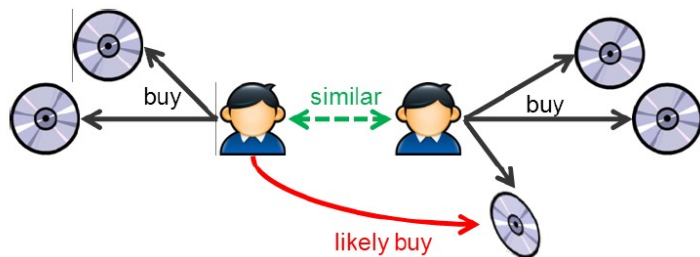


By Chris Duckett | October 19, 2017 -- 00:44 GMT (08:44 GMT+08:00) | Topic: Innovation

Is AlphaGo Zero a scientific breakthrough – step towards AGI?



Special task



IRIS

AI use cases

- **Supervised learning**

- Price estimation
- Direct target marketing
- Credit scoring

- **Unsupervised learning**

- Customer segmentation
- Fraud detection
- Campaign/promotion

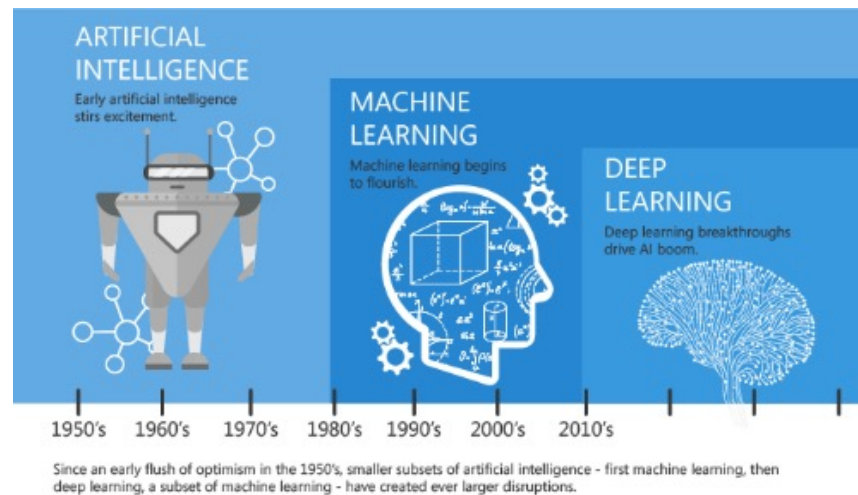
- **Reinforcement learning**

- Optimization & autonomous

- **Special tasks**

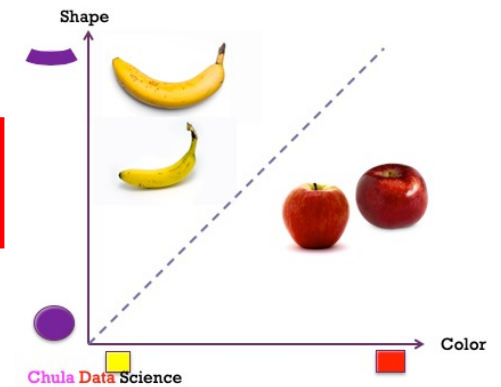
- Product recommendation
- Inventory forecasting
- Customer forecasting
- Etc.

Deep Learning (DL)



Handcrafted features

Age	Income	Gender	Province	Purchase
25	25,000	Female	Bangkok	Yes
35	50,000	Female	Nontaburi	Yes
32	35,000	Male	Bangkok	No



Can we still tell the features (columns)?



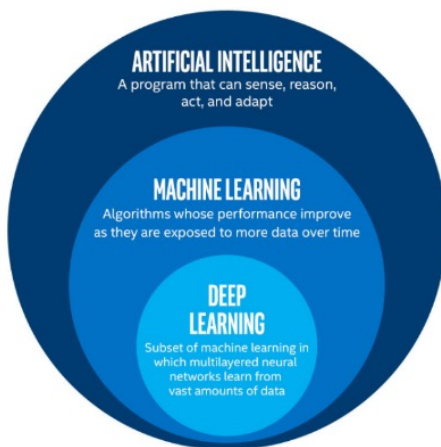
What is Deep Learning (DL)?



Part of the **machine learning** field of learning representations of data. Exceptional effective at learning patterns.

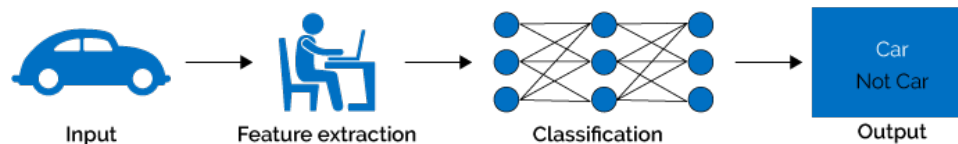


Utilizes learning algorithms that derive meaning out of data by using a **hierarchy** of multiple layers that **mimic the neural networks of our brain**.

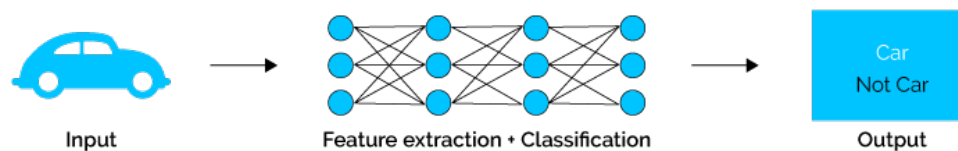


Costs lots of time

Machine Learning



Deep Learning



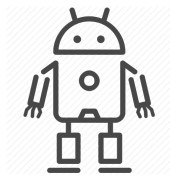
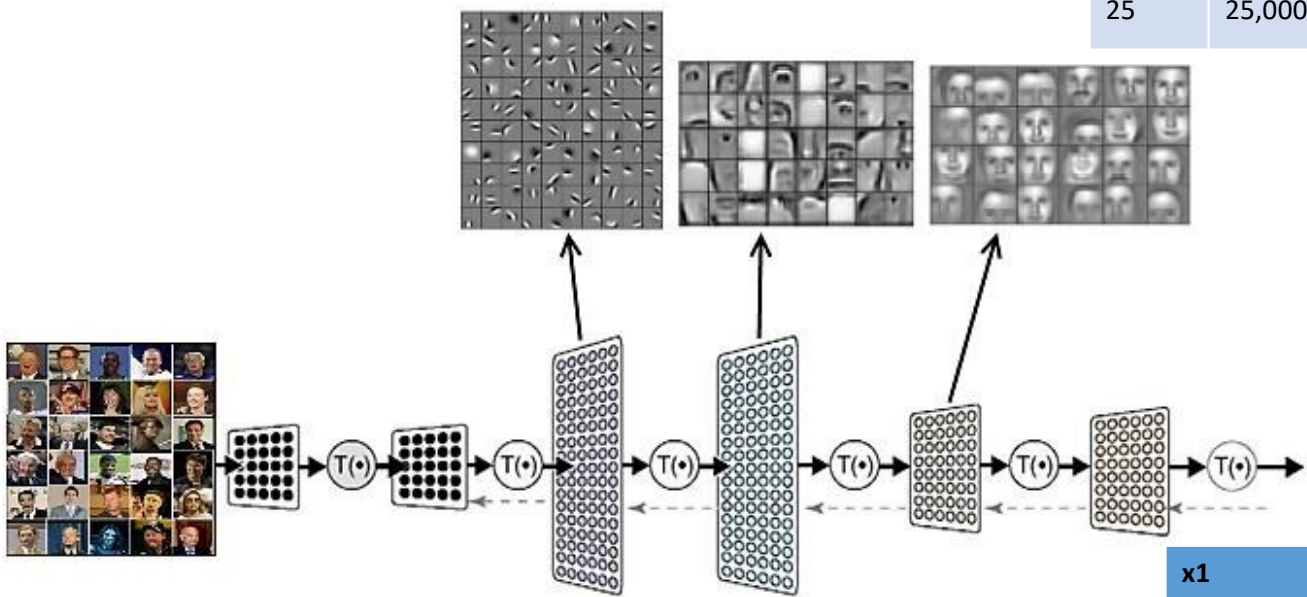
Deep Learning – Basics (cont.)

What did it learn?

A deep neural network consists of a **hierarchy of layers**, whereby each layer **transforms the input data** into more abstract representations (e.g., edge -> nose -> face). The output layer combines those features to make predictions.



Age	Income	Gender	Province	Corona
25	25,000	Female	Bangkok	Yes

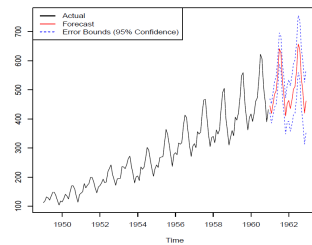


x1	x2	x3	x4	Corona
0.7	0.2	-0.5	-0.1	Yes

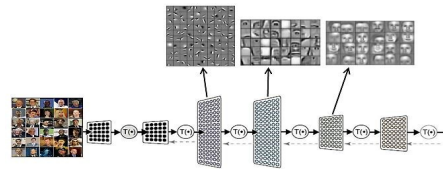
Deep Learning Application



**Speech
Recognition**



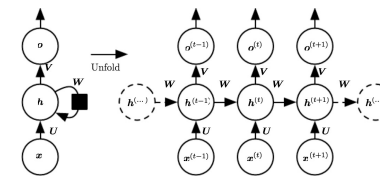
**Computer
Vision**



CNN



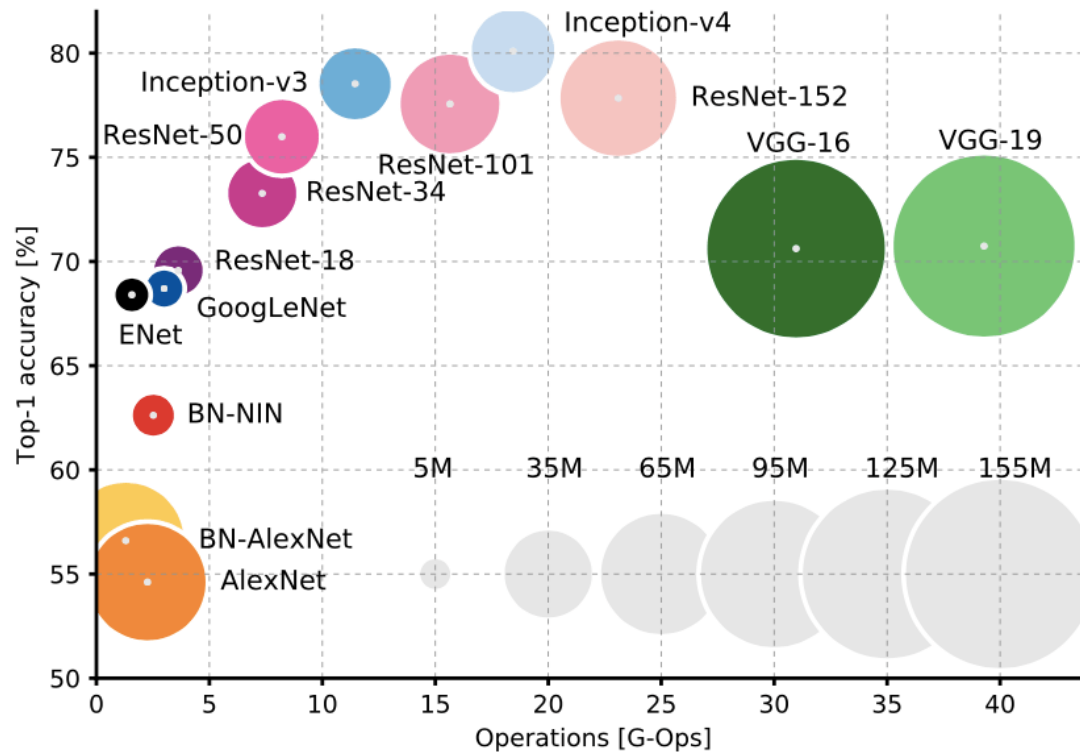
**Natural Language
Processing**



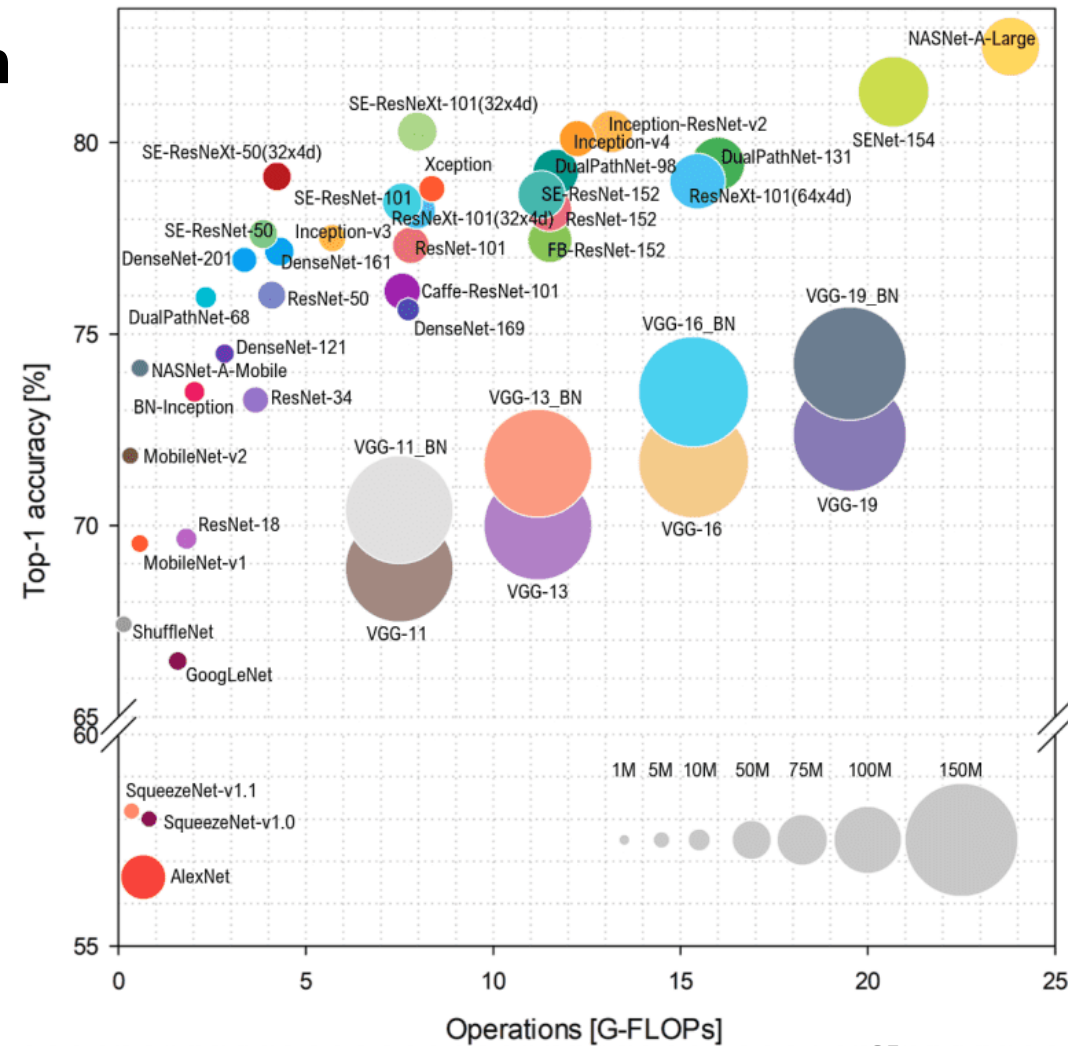
RNN (LSTM)

$$\text{Spend} = 500 + 10 * \text{Age} + 20 * \text{Income1K}$$

SOTA of Image Classification



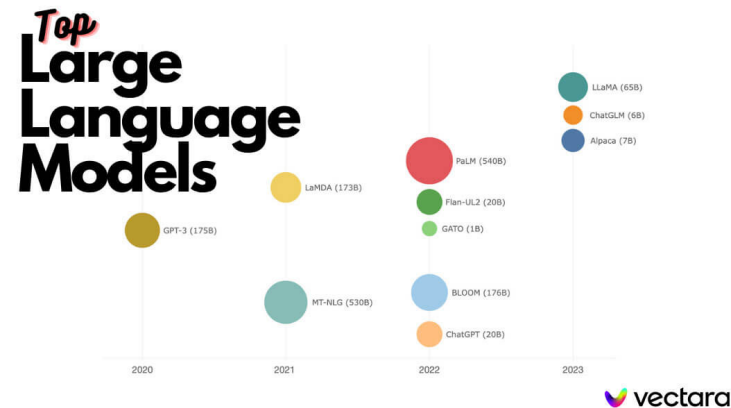
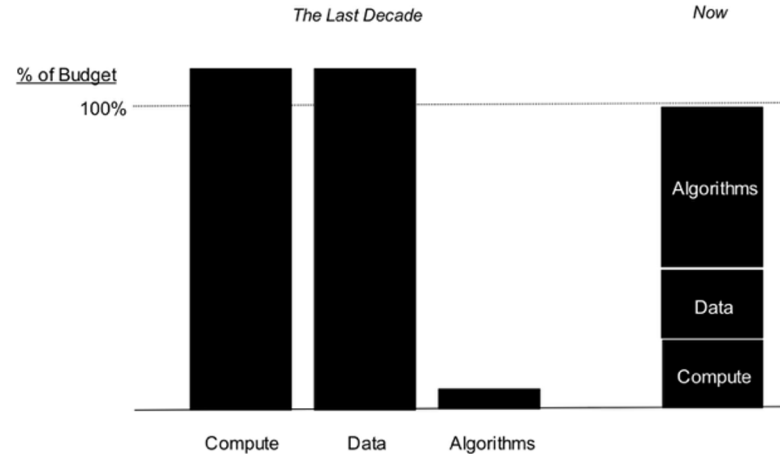
https://blog.csdn.net/qg_34216467/article/details/83061692



<https://theaisummer.com/cnn-architectures/>

Why now

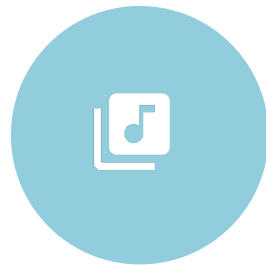
- Neural Networks has been around since 1990s
- **Big data** – DNN can take advantage of large amounts of data better than other models
- **GPU** – Enable training bigger models possible
- **Deep (pretrained models)** – Easier to avoid bad local minima when the model is large



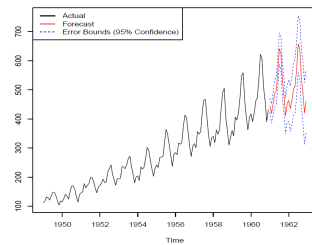
Based on the slide of Aj.Ekapol

<http://www.kdnuggets.com/2017/06/practical-guide-machine-learning-understand-differentiate-apply.html>

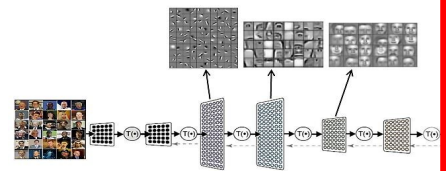
Deep Learning Application



**Speech
Recognition**



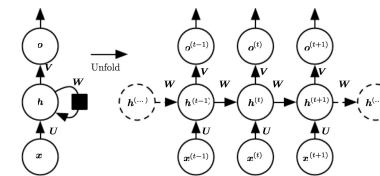
**Computer
Vision**



CNN



**Natural Language
Processing**



RNN (LSTM)

Type of image tasks

Semantic Segmentation



GRASS, CAT,
TREE, SKY

No objects, just pixels

Classification + Localization



CAT

Single Object

Object Detection



DOG, DOG, CAT

Multiple Object

Instance Segmentation



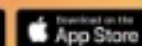
DOG, DOG, CAT

This image is CC0 public domain

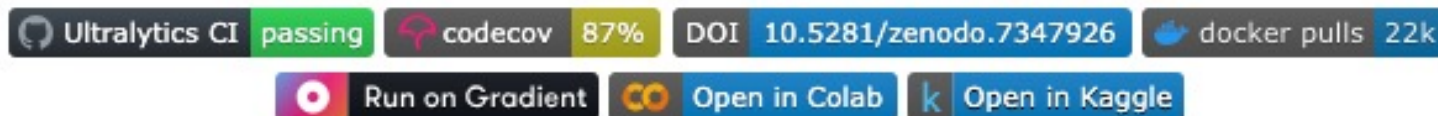
<https://github.com/ultralytics/ultralytics>



DOWNLOAD THE APP



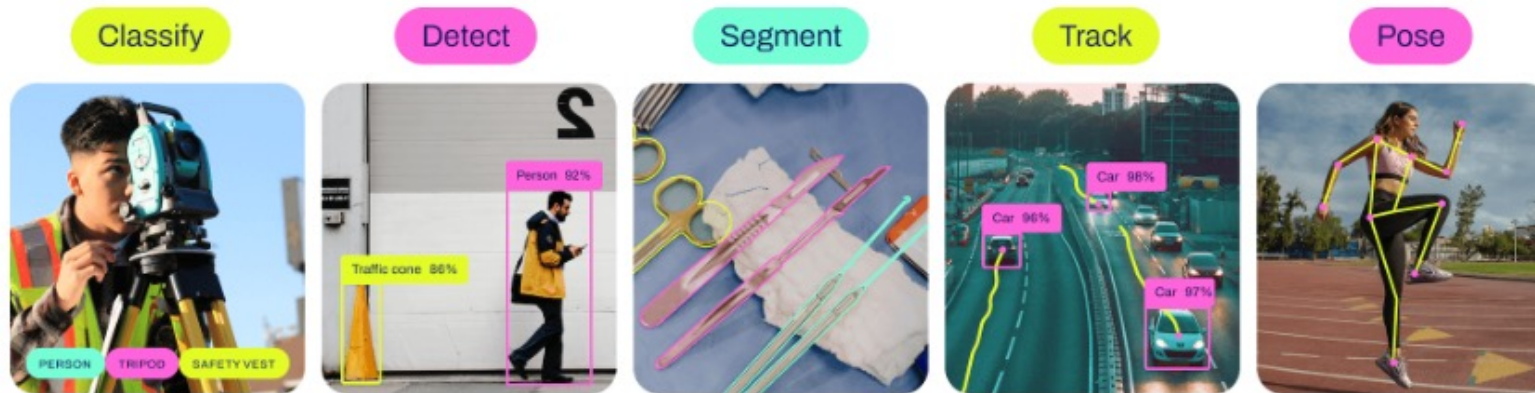
[English](#) | [简体中文](#)



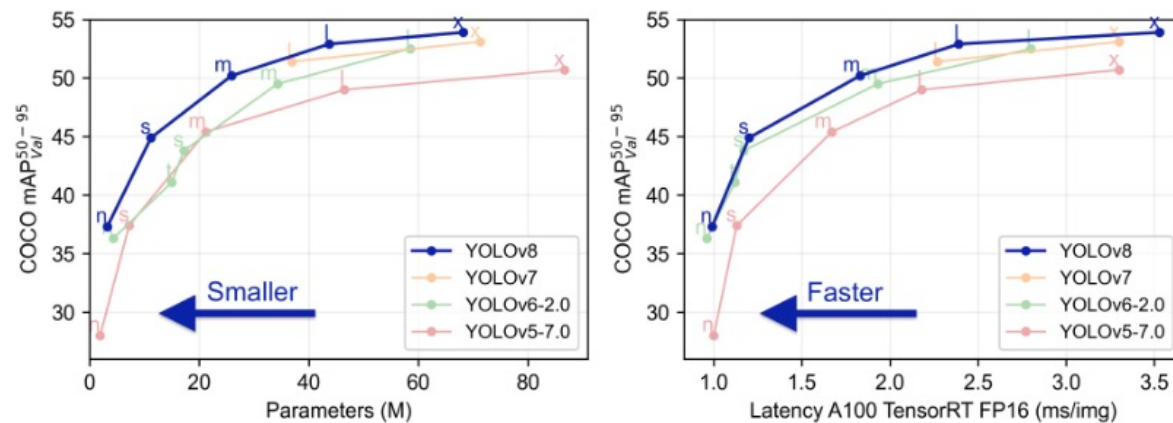
Ultralytics YOLOv8 is a cutting-edge, state-of-the-art (SOTA) model that builds upon the success of previous YOLO versions and introduces new features and improvements to further boost performance and flexibility. YOLOv8 is designed to be fast, accurate, and easy to use, making it an excellent choice for a wide range of object detection and tracking, instance segmentation, image classification and pose estimation tasks.

Models

YOLOv8 [Detect](#), [Segment](#) and [Pose](#) models pretrained on the [COCO](#) dataset are available here, as well as YOLOv8 [Classify](#) models pretrained on the [ImageNet](#) dataset. [Track](#) mode is available for all Detect, Segment and Pose models.



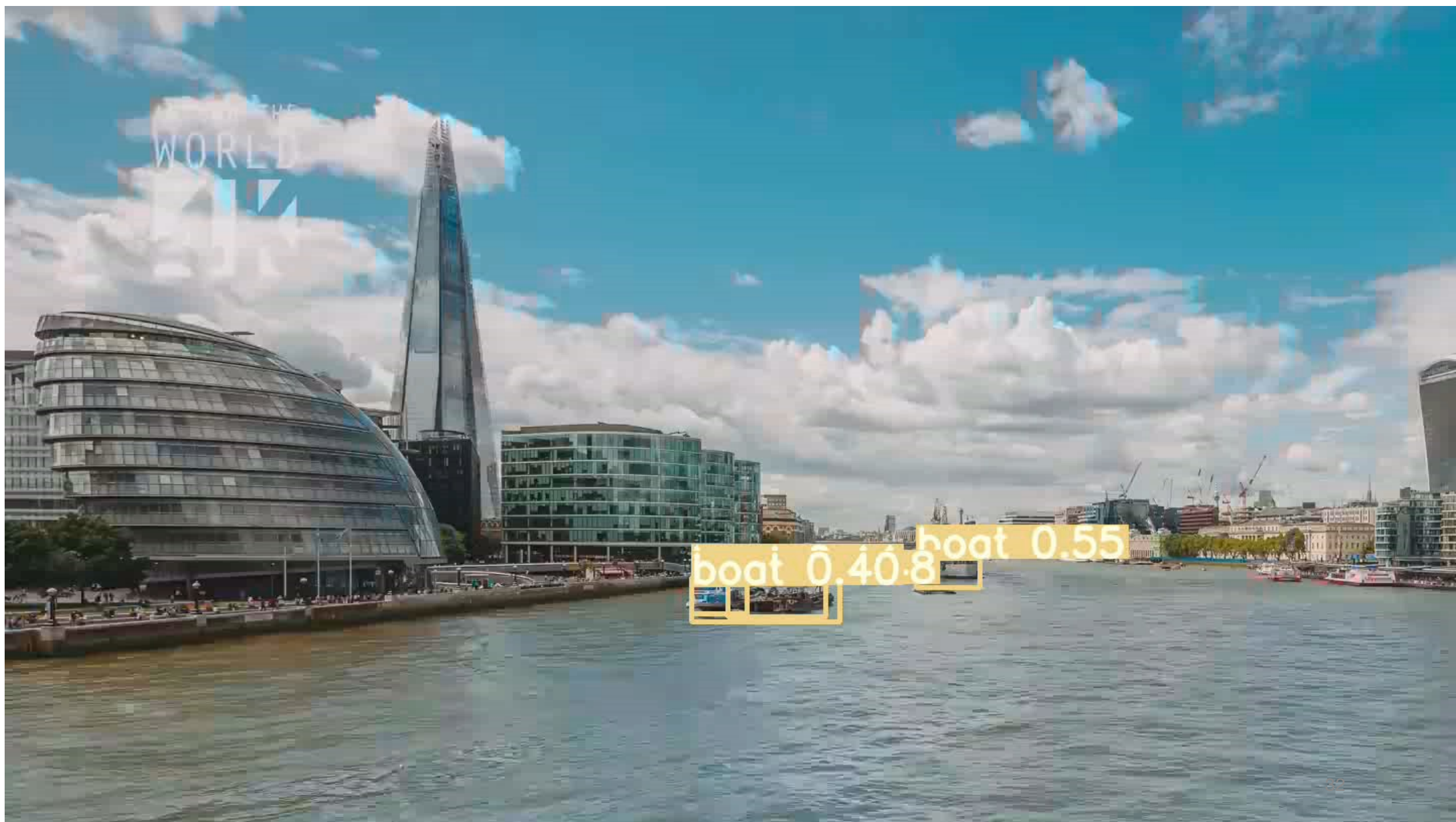
All [Models](#) download automatically from the latest Ultralytics [release](#) on first use.



PEOPLE COUNTING



<https://www.hikvision.com/th/newsroom/latest-news/2019/people-counting-for-increased-retail-success/>



In Frame: 3
Identified: 3



www.youtube.com/workpointofficial

traffic sign 0.54

traffic sign 0.12

traffic sign 0.39

billboard 0.33

car 0.76

car 0.89

car 0.86

car 0.87

car 0.1

car 0.94

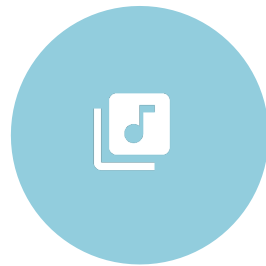
traffic cone 0.67

car 0.93

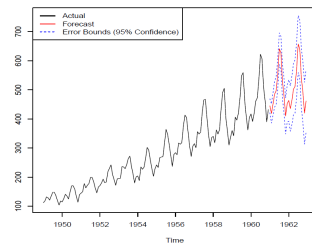
car 0.95



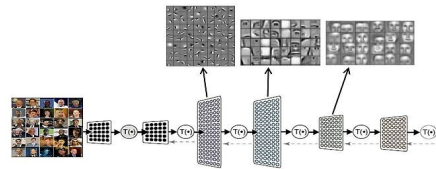
Deep Learning Application



**Speech
Recognition**



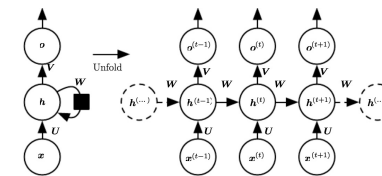
**Computer
Vision**



CNN



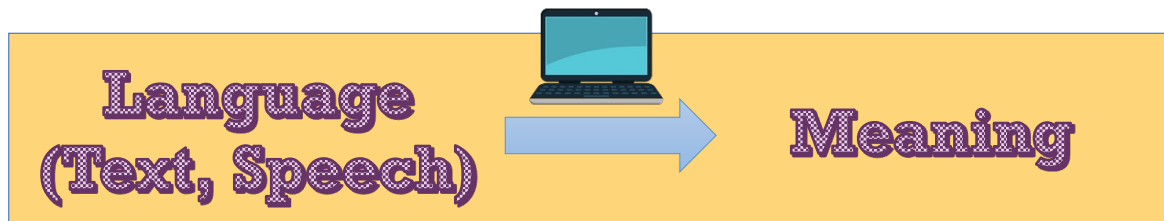
**Natural Language
Processing**



RNN (LSTM)

Introduction: Natural Language Processing (NLP)

- Subfield of AI
- GOAL:



- Bridge the gap between **how people communicate** and **what machines understand** in order to perform useful tasks, e.g.
 - Making appointments, buying things, question answering, etc.

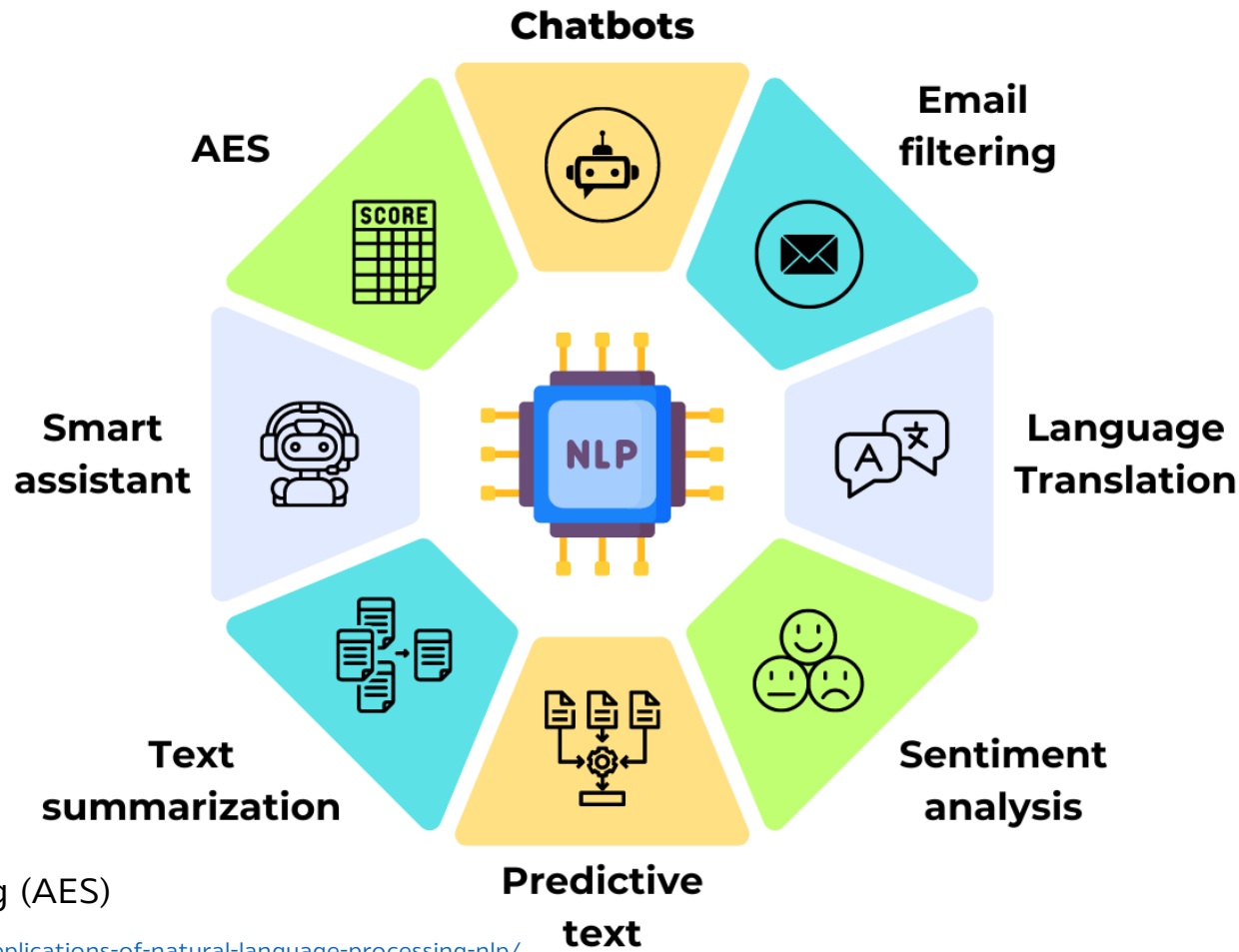


Goal: intelligent processing of human languages

- **Not just** string matching

Top 8 Applications of Natural Language Processing (NLP)

Applications of Natural Language Processing



<https://eastgate-software.com/top-8-applications-of-natural-language-processing-nlp/>

ChatGPT (Generative AI)

The screenshot displays the ChatGPT web interface. On the left is a sidebar with a 'New chat' button and a list of previous conversations. The main area shows a chat history with a conversation titled 'ขอรายละเอียดบริษัท BJC Thailand'. The chat includes a user query in Thai and a detailed response from ChatGPT. The response describes BJC Thailand as a retail and food producer, lists its four main business groups (hypermarkets, food production, hospitals, and real estate development), and provides sales and employee statistics for 2020. A second user query asks about 'Big C', and ChatGPT responds by identifying it as a hypermarket under BJC Thailand, listing its product categories, and providing further sales and branch statistics for 2020. At the bottom, there is a text input field with the placeholder 'Send a message.' and buttons for 'Stop generating' and 'Regenerate response'.

Chat History:

- User:** ขอรายละเอียดบริษัท BJC Thailand
- ChatGPT:** BJC Thailand เป็นบริษัทค้าปลีกและผู้ผลิตอาหารและเครื่องใช้ในบ้านในประเทศไทย ก่อตั้งขึ้น ในปี พ.ศ. 2517 (1974) โดยมีกลุ่มธุรกิจที่เกี่ยวข้องกับกลุ่มธุรกิจ Charoen Pokphand Group เป็นผู้ถือหุ้นส่วนสูงสุดในบริษัท

BJC มีธุรกิจหลัก 4 กลุ่ม ได้แก่
 1. ธุรกิจค้าปลีก: ซูเปอร์มาร์เก็ต, ร้านสะดวกซื้อ, ร้านค้าออนไลน์ เช่น Big C, Tesco Lotus, ฯลฯ
 2. ธุรกิจผลิตอาหารและเครื่องดื่ม: ผลิตและจำหน่ายอาหารและเครื่องดื่ม เช่น น้ำอัดลม, น้ำดื่ม, น้ำผลไม้, อาหารสำเร็จรูป เช่น Oishi, Green Spot, Masita, Malee, Del Monte, UFC
 3. ธุรกิจโรงแรมและสปา: มีโรงแรมและสปาในหลายจังหวัดของไทย รวมถึงเกาะสมุยและเขาใหญ่ ยังมีธุรกิจเกี่ยวกับการจัดงานประชุม และงานอีเวนต์ต่างๆ
 4. ธุรกิจที่เกี่ยวข้องกับการพัฒนาอสังหาริมทรัพย์: เช่น การลงทุนในโครงการอสังหาริมทรัพย์ที่เกี่ยวข้องกับธุรกิจของบริษัท
ในปี 2563 (2020) BJC มีรายได้จากการขายสุทธิ 408,695 ล้านบาท และมีพนักงานประมาณ 140,000 คน โดยมีการขยายธุรกิจไปยังต่างประเทศ
- User:** แล้วเกี่ยวกับ Big C ยังไง?
- ChatGPT:** Big C เป็นธุรกิจค้าปลีกของ BJC Thailand ซึ่งเป็นซูเปอร์มาร์เก็ตในลักษณะของ "hypermarket" โดยจำหน่ายสินค้าหลากหลายประเภท เช่น เครื่องใช้ไฟฟ้า, เครื่องดนตรี, เครื่องสำอาง, เครื่องแต่งกาย, ผลิตภัณฑ์เสริมอาหาร, ผลิตภัณฑ์เพื่อสุขภาพ, ผลิตภัณฑ์อาหารสัตว์, ผลิตภัณฑ์ไอที, รถยนต์, รถจักรยานยนต์, อุปกรณ์กีฬา, อุปกรณ์เดินทาง และสินค้าอื่นๆ อีกมากมาย

โดย Big C มีสาขาทั้งในประเทศไทยและต่างประเทศ และมีการขยายธุรกิจออนไลน์ของบริการจัดส่งสินค้าด้วย

ในปี 2563 (2020) ธุรกิจค้าปลีกของ BJC Thailand รายการ Big C มีรายได้จากการขายสุทธิ 165,464 ล้านบาท และมีจำนวนสาขาทั้งหมด 147 สาขา โดยมีรายการค้าปลีกอื่นๆ เช่น Tesco Lotus, Makro, และธุรกิจค้าปลีกออนไลน์อย่าง Lazada และ JD Central อยู่ในกลุ่มธุรกิจค้าปลีกของ BJC Thailand ด้วย

ChatGPT^[a] is an [artificial intelligence](#) (AI) [chatbot](#) developed by [OpenAI](#) and released in November 2022. It is built on top of OpenAI's [GPT-3.5](#) and [GPT-4 foundational large language models](#) (LLMs) and has been [fine-tuned](#) (an approach to [transfer learning](#)) using both [supervised](#) and [reinforcement learning](#) techniques.

Tools for Thai NLP



PyThaiNLP

python 3.6 pypi v2.1.2 downloads/month 15k License Apache 2.0 license scan passing build passing build passing
code quality A coverage 91% Launch Quick Start Guide on Google Colab DOI 10.5281/zenodo.3595968

Thai Natural Language Processing in Python.

PyThaiNLP is a Python package for text processing and linguistic analysis, similar to `nltk` but with focus on Thai language.

News

We are conducting a 2-minute survey to know more about your experience using the library and your expectations regarding what the library should be able to do. Take part in this survey: <https://forms.gle/aLdSHnvkNuk5CFy9>

This is a document for development branch (post 2.1). Things will break.

WangchanBERTa โมเดลประมวลผลภาษาไทยที่ใหญ่และก้าวหน้าที่สุดในขณะนี้



VISTEC-depa AI Research Institute of Thailand

Follow

Jan 24 · 5 min read

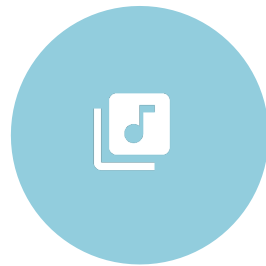


เปิดให้ทุกคนใช้ฟรีโดยAIRsearch.in.th และVISTEC ภายใต้สัญญาอนุญาต CC-BY-SA 4.0

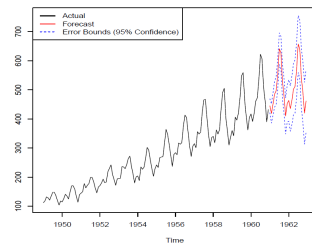


IRIS

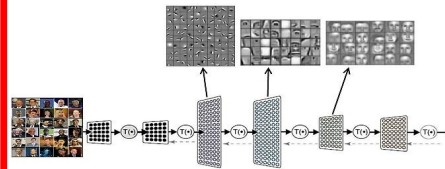
Deep Learning Application



**Speech
Recognition**



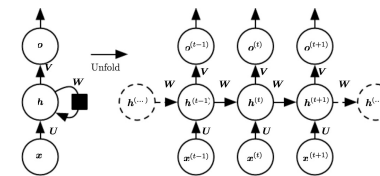
**Computer
Vision**



CNN



**Natural Language
Processing**



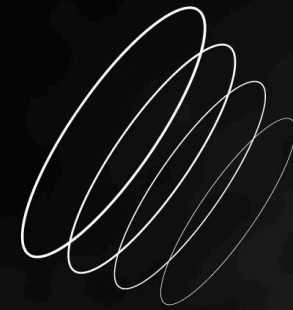
RNN (LSTM)

Thonburian Whisper



IRIS

 GOWAJEE



GOWAJEE API
LAUNCH

[Dashboard](#)[AI Marketplace](#)[Library](#)[Sign in](#)[AI Marketplace](#) > [Speech To Text](#)

Speech To Text

WER: 10.6%

CER: 4.7%

DER: 13.5%

TH

API Document

[Add to Library](#)

AI can transcribe speech into text, supporting multiple file types such as mp3, wav, and flac.

Publisher: Gowajee

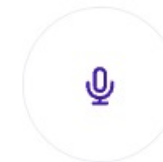
[Let's try demo](#)[How to call the API?](#)[Full evaluation](#)

We utilized a model from our partner, Gowajee. The model is trained by using over 1,000 hours of annotated data collected online by our partner from various sources. This model can perform in general topics but specializes in call center. The model could be worsened if audio contains code switching, low-quality speech, and overlapping speech data. The model's performance in the term of transcribe speech was evaluated on around 60 hours of speech data, which was collected from the same source as the training data. For diarization, it was evaluated on artificial conversations created from audio pools containing around 50 speakers. Each speaker appears only once in a conversation, ensuring there is no overlap between speakers across conversations.

For more information, visit [Guide Book](#)

[Pricing](#)

Let's try demo

**Record audio**

You can record audio up to 1 minute

or

[Click to upload](#) or drag and drop

<https://console.visai.ai/ai-marketplace/ready-to-use/c7ba44f3-0de6-4457-8743-75438e3eac60/speech-to-text>

Thonburian Whisper Demo 🇹🇭

Thonburian Whisper

Transcribe Audio

Transcribe YouTube

Transcribe long-form microphone or audio inputs with the click of a button! Demo uses the fine-tuned checkpoint [biodatlab/whisper-th-medium-combined](#) and 🧠 Transformers to transcribe audio files of arbitrary l

Microphone Input



Record

Default - MacBook ...

Transcription Output

Audio File Upload



Drop Audio Here

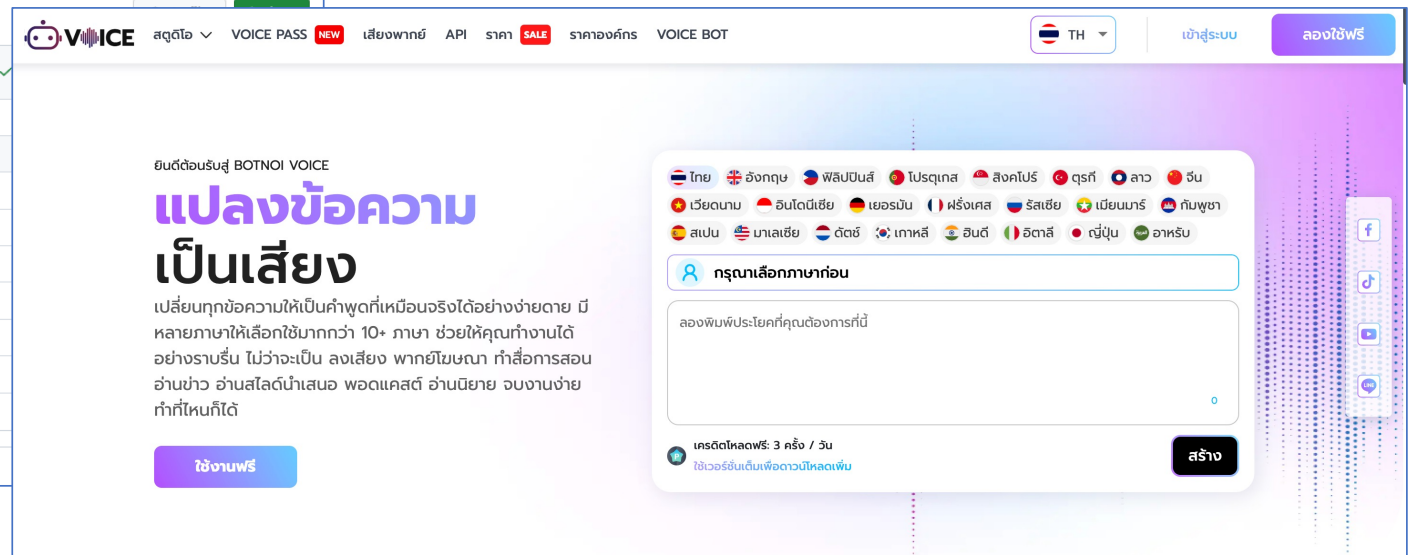
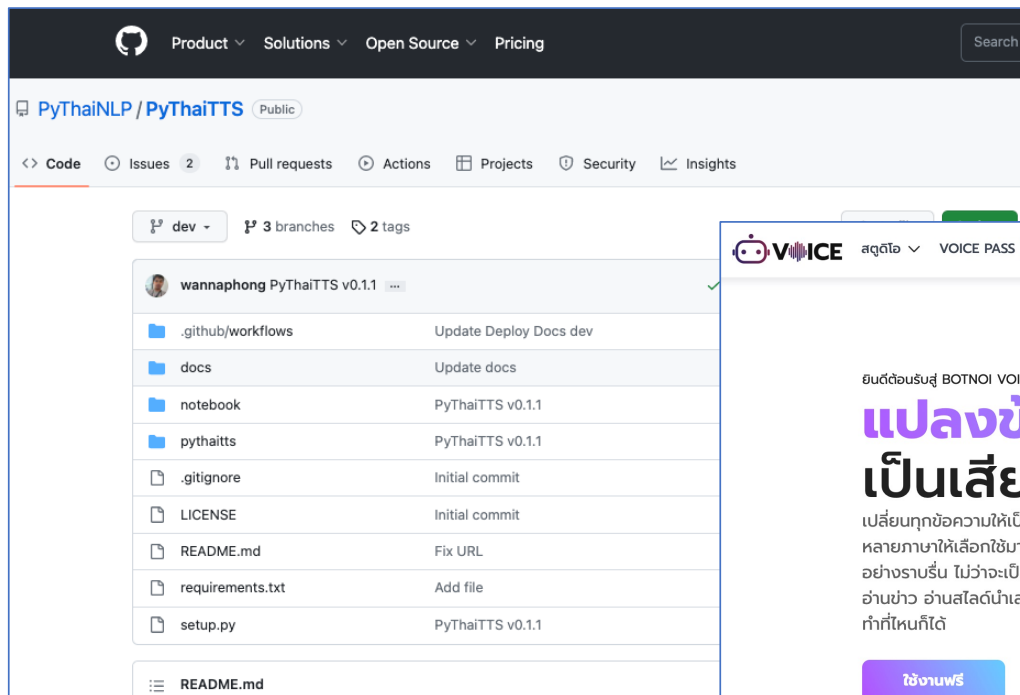
- or -

Click to Upload

https://huggingface.co/spaces/biodatlab/whisper-thai-demo?fbclid=IwZXh0bgNhZW0CMTEAAR0LKUeVwIMoRPCpepgMA00j0jtvVfLBEWetTzZQshg8a7_oq1UARafqC4_aem_hgR19w4x48qUtGcwUFRiiQ

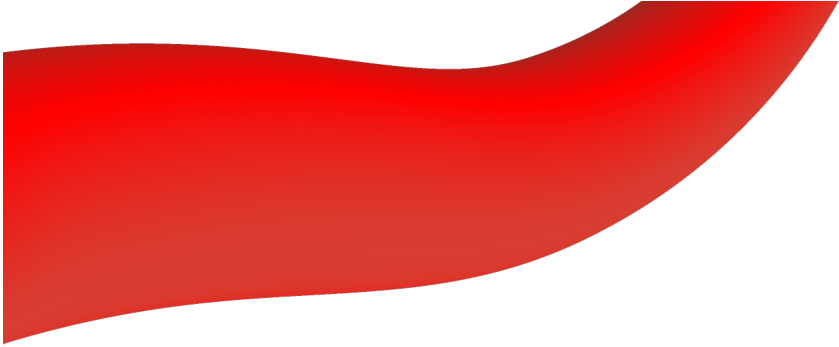
Transcribe

Text-to-Speech (TTS)

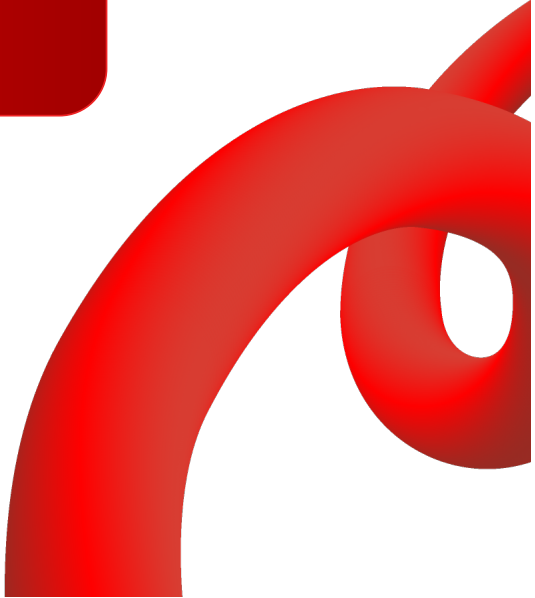


<https://github.com/PyThaiNLP/PyThaiTTS>

<https://voice.botnoi.ai/>



Generative AI



IRIS

What is Generative AI?

Generative artificial intelligence (AI) is a type of AI **that can create new content, such as text, images, music, and videos**. It uses machine learning models to learn from its training data to “generate” new statistically probable outputs.

Artificial Intelligence

Is the field of study

Machine Learning

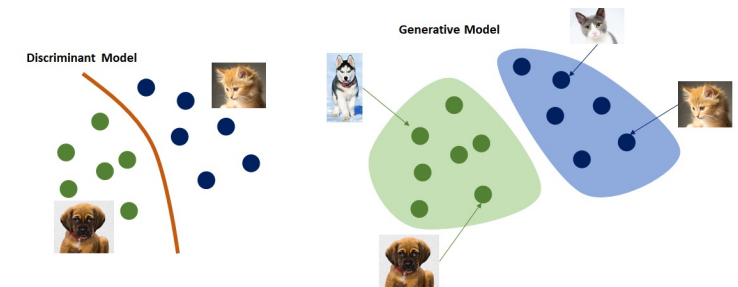
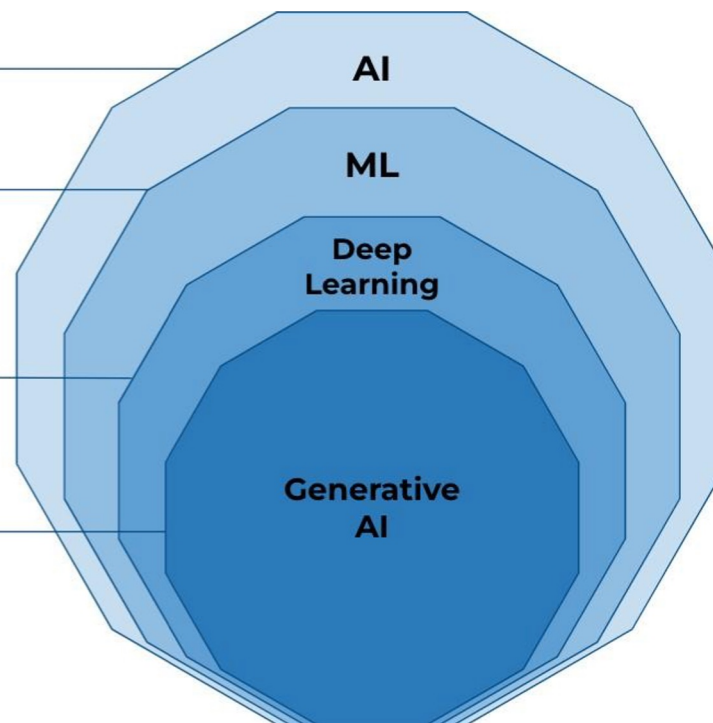
Is a branch of AI that focus on the creation of intelligent machines that learn from data. Another very well know branch inside AI is **Optimization**.

Deep Learning

Is a subset of Machine Learning methods, based on **Artificial Neural Networks**.
Examples: CNNs, RNNs

Generative AI

A type of ANNs that generate data that is similar to the data it was trained on.
Examples: GANs, LLMs



What can Gen AI create?

- Text
- Image
- Video
- Code
- Multimodal

Example of Gen AI: ChatGPT

- ChatGPT was launched by OpenAI on 30 Nov 2022.
- ChatGPT is a large language model (LLM) for conversational AI applications.
- Generates human-like text and performs NLP tasks.
- Scalable and flexible for various use cases.
- ChatGPT didn't enclose the details.

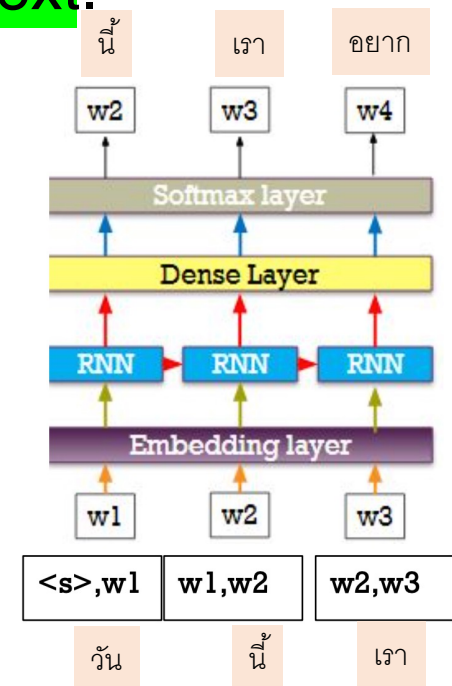


Language Model (LM)

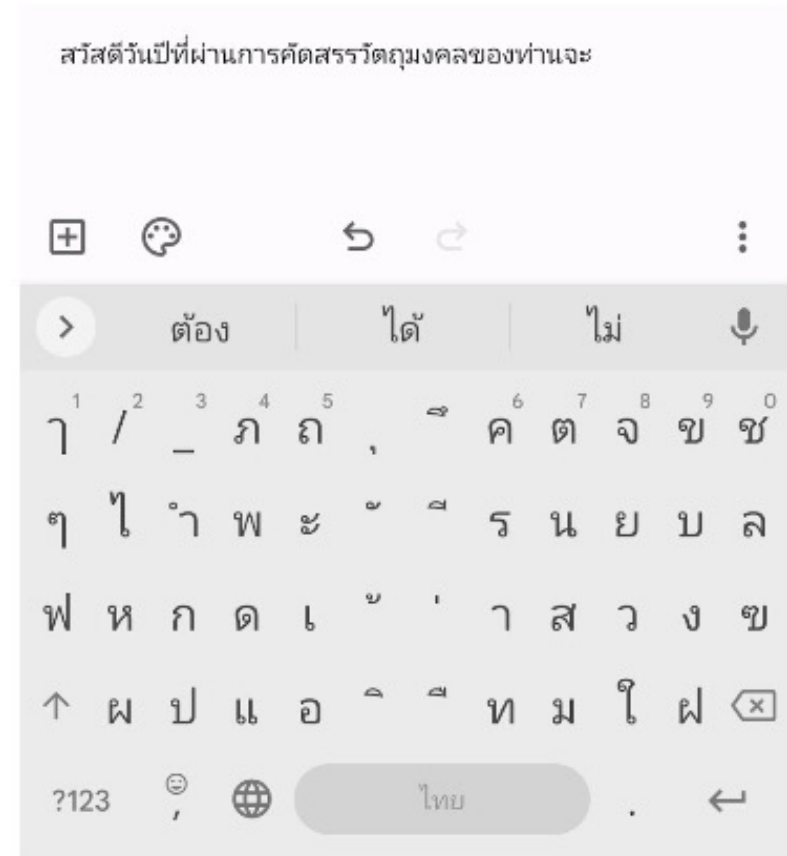
- It is the model that aims to **predict next word** based on the given previous words.
- So, the model can understand **grammar & context**.

วันนี้เราอยากจะกินข้าวมัน_____

วัน นี้ เรา อยาก จะ กิน ข้าว มัน



สร้างข้อความด้วย AI

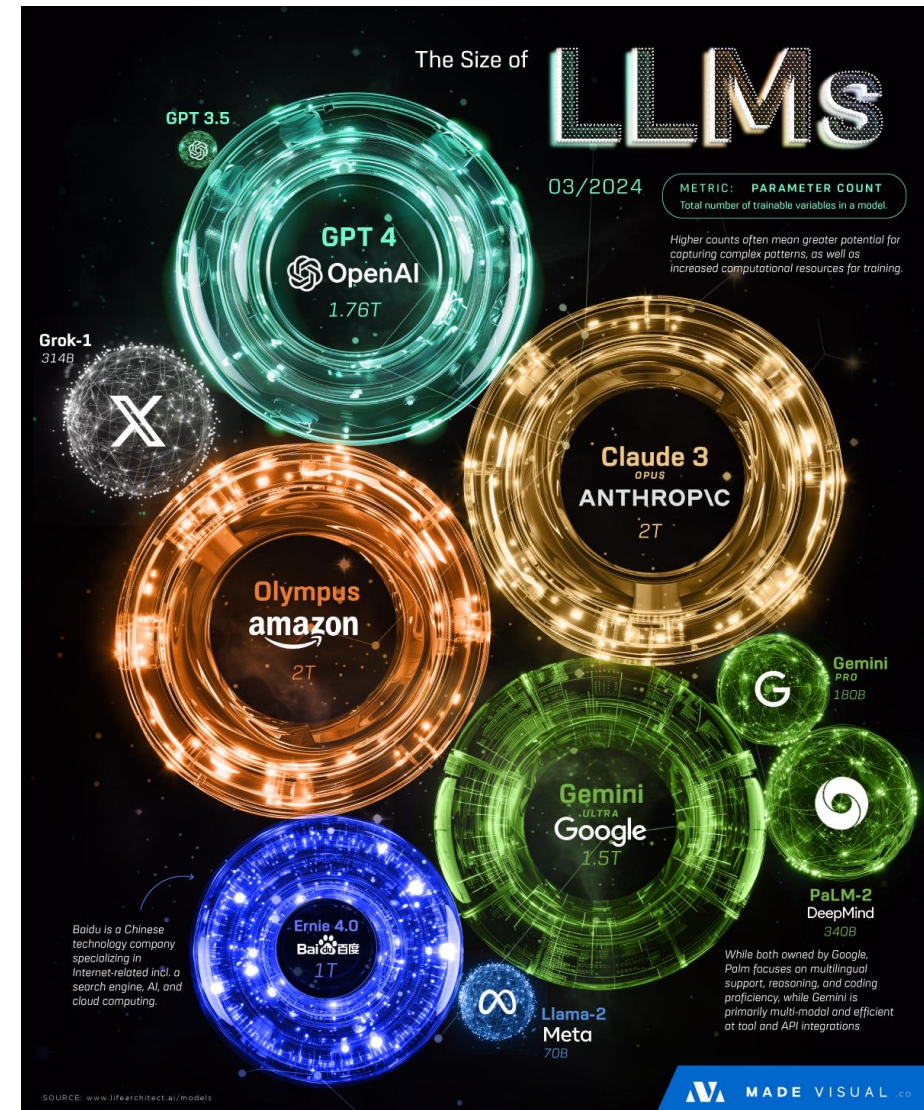


Large + LM = LLM

Model	Year	Parameters (approx)	Tokens Seen (approx)
GPT-1	2018	117M	~0.1B
GPT-2	2019	1.5B	~10B
GPT-3	2020	175B	~300B
GPT-4	2023	~1.7T (MoE est.)	~13T
GPT-5	2025	300B	~114T

Amazon now develops its own family of **foundation models** called **Amazon Nova** (launched late 2024):

- **Nova Micro** → lightweight text-only model (fast, low cost)
- **Nova Lite** → multimodal (text, images, video), efficiency-focused
- **Nova Pro** → stronger multimodal, balance between accuracy & speed
- **Nova Premier** → top-tier multimodal model (teacher model for distillation)
- **Nova Canvas** → image generation
- **Nova Reel** → video generation
- **Nova Sonic** → speech-to-speech (voice)
- **Nova Act** → agent model that can operate tools and browsers



Popular LLM

Currently, large language models are very popular and there are many models available for use. They are categorized by their application as follows:

1. Open-source models: [Llama-4](#), [Mistral](#), [DeepSeek-V3](#), [Gemma-3](#), [Qwen-3](#)
2. Closed-source models: [GPT-5](#), [Claude](#), [Gemini](#)
3. Specific models: [Typhoon-2.5](#), [SEA-LION](#)

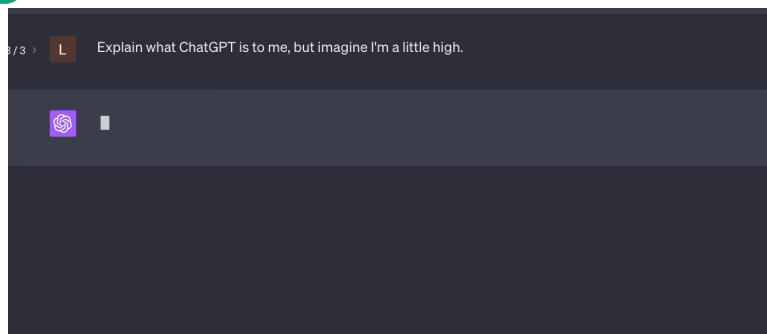


LLM Use Cases

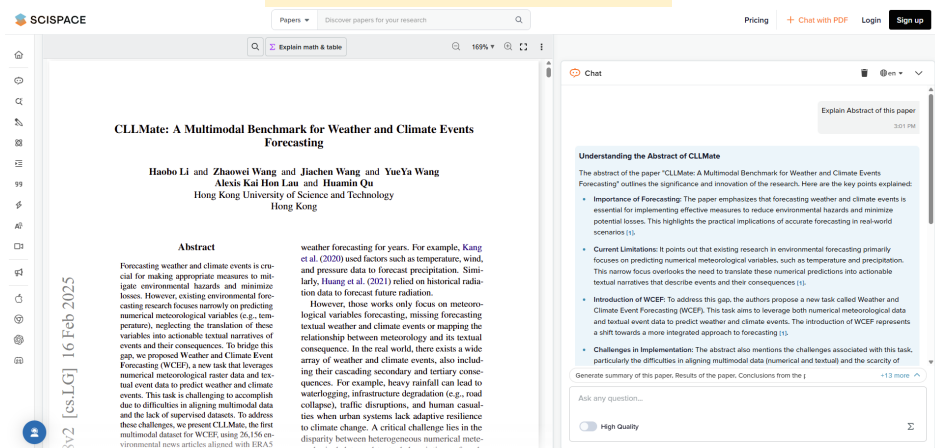


ChatGPT

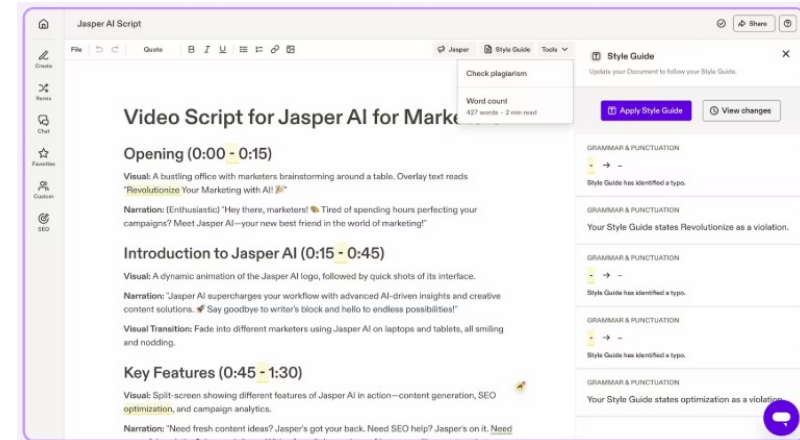
1) Assistant Chatbot



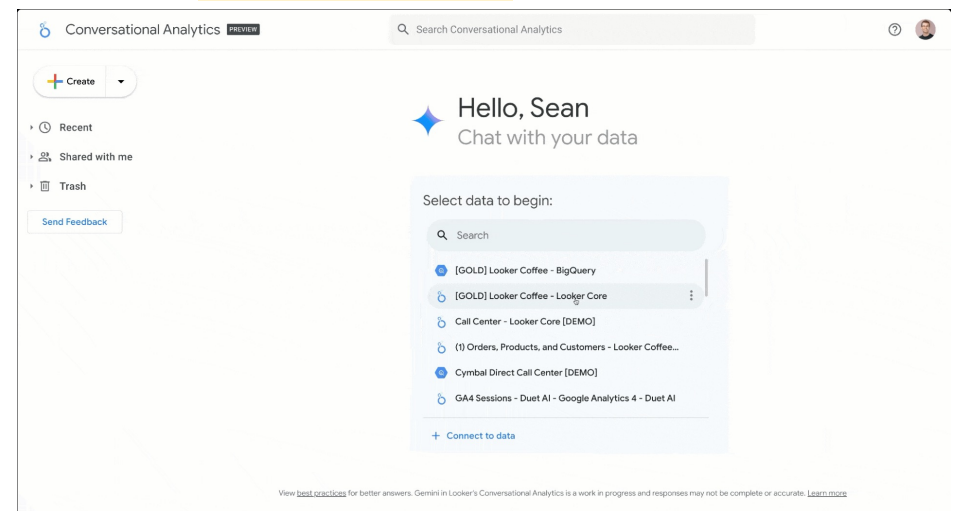
2) Text Summarization



3) Content Creation



4) Data Analytics



Prompt Engineering Techniques



Zero-shot
prompting



Few-shot prompting
or in-context
learning



Chain-of-thought
prompting



Tree-of-thought
prompting



Iterative
refinement



Feedback
loops



Prompt
chaining



Role-playing



Maieutic
prompting



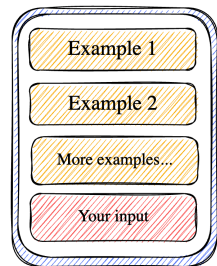
Complexity-based
prompting

servicenow

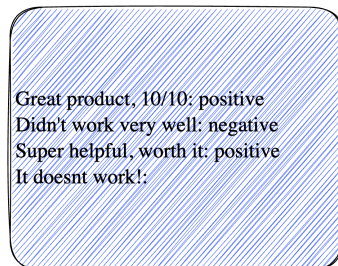
<https://www.servicenow.com/ai/what-is-prompt-engineering.html>

1) Few-Shot Prompting

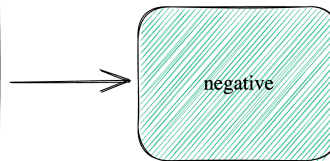
A Few Shot Prompt



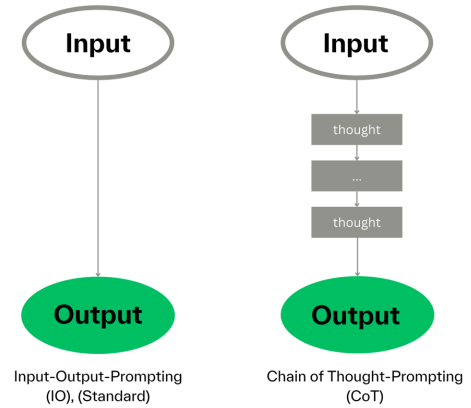
Example



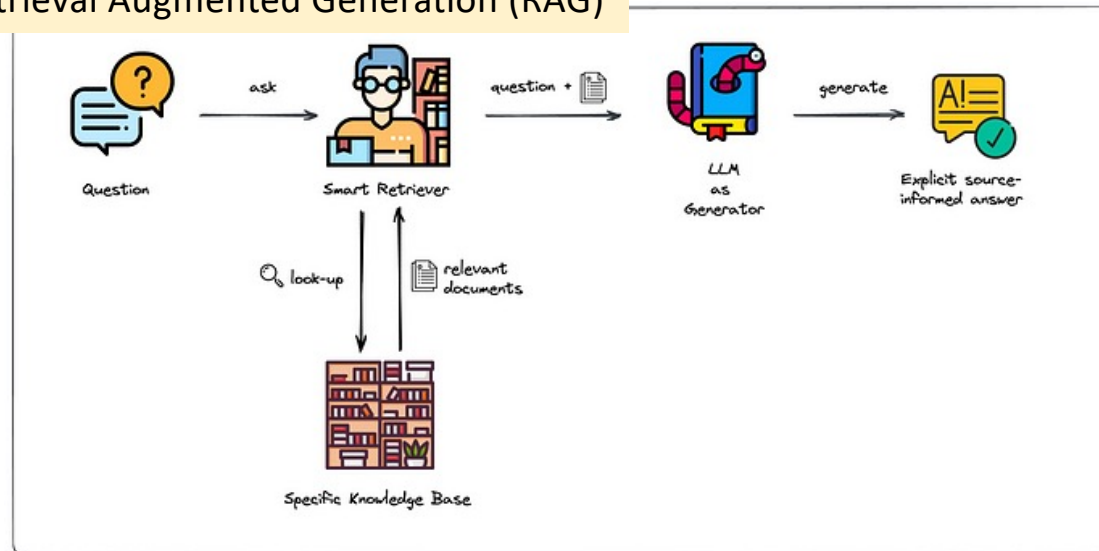
Model Output



2) Chain-of-Thoughts (CoT)

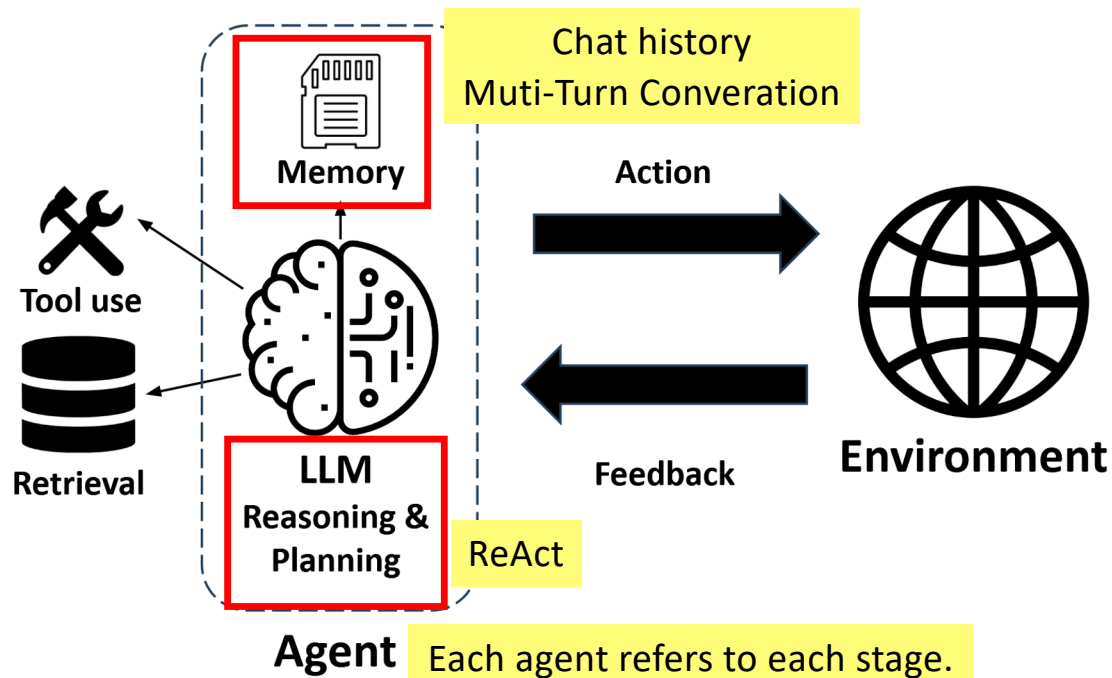


3) Retrieval Augmented Generation (RAG)

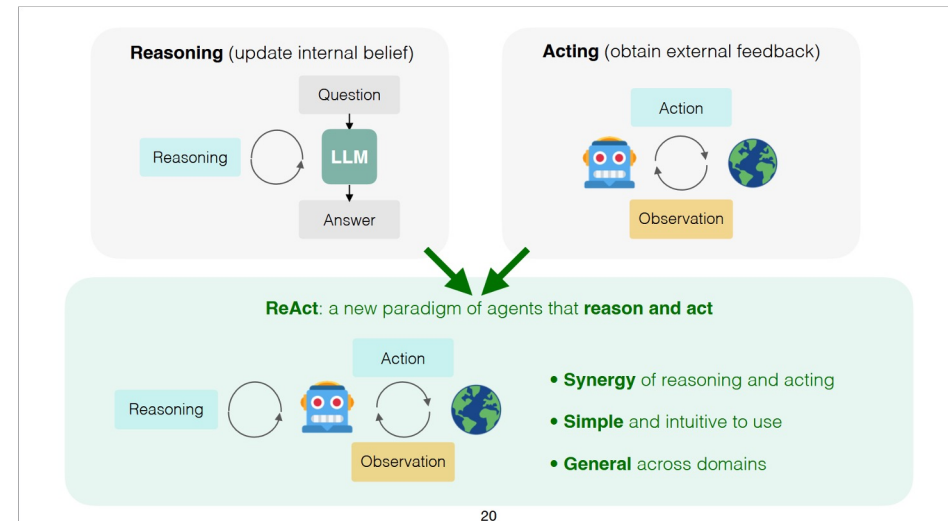


Agentic LLMs

An agentic LLM is a large language model (LLM) that operates as an autonomous agent, capable of performing tasks, making decisions, and interacting with external tools or APIs to achieve the specified goals.



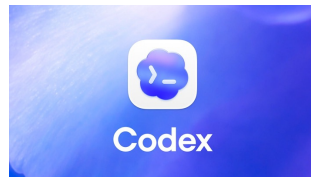
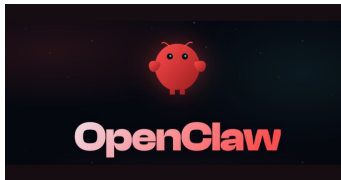
Reasoning: ReAct = Reasoning + Acting



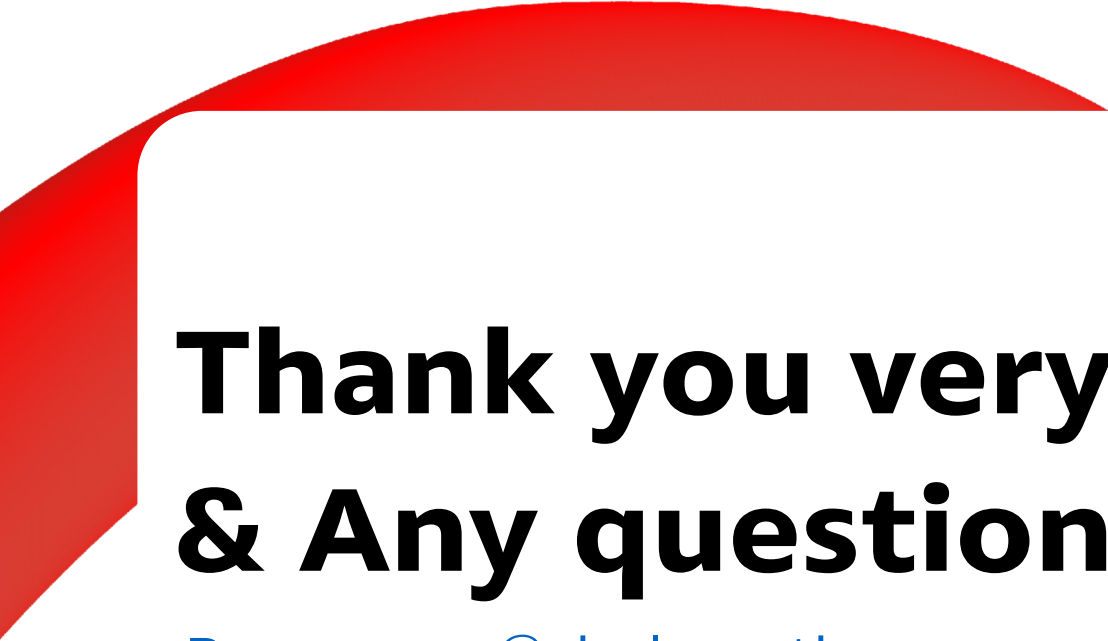
20

AI Agentic Tools in 2026

- How to automate?
 - Use tools
 - Execute actions
 - Observe results
 - Reflect
 - Continue autonomously



Tool	Company	Primary Focus
Claude Code	Anthropic	Autonomous coding, codebase understanding, debugging, refactoring
Codex	OpenAI	Software engineering agent, coding tasks, testing, code generation
OpenClaw	Open Source	Computer-use agent, workflow automation, personal assistant tasks
Gemini CLI	Google DeepMind	Agentic coding and terminal-based development workflows
Cursor	Anysphere	AI-native IDE with agentic coding capabilities
Devin	Cognition	Autonomous software engineering agent
GitHub Copilot	GitHub	AI-assisted coding with increasing agentic capabilities

A large red decorative shape on the left side of the slide, consisting of a curved top and a vertical edge.

Thank you very much & Any questions

Peerapon.v@chula.ac.th